



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Trade Secret Protection and the Integration of Information Within Firms

Ionela Andreicovici; , Sara Bormann, Katharina Hombach

To cite this article:

Ionela Andreicovici; , Sara Bormann, Katharina Hombach (2025) Trade Secret Protection and the Integration of Information Within Firms. Management Science 71(2):1213-1237. <https://doi.org/10.1287/mnsc.2021.03484>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2024, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Trade Secret Protection and the Integration of Information Within Firms

Ionela Andreicovici,^a Sara Bormann,^{b,*} Katharina Hombach^b

^aDepartment of Accounting, Frankfurt School of Finance & Management, 60322 Frankfurt am Main, Germany; ^bDepartment of Accounting, Goethe University Frankfurt, 60323 Frankfurt am Main, Germany

*Corresponding author

Contact: i.andreicovici@fs.de,  <https://orcid.org/0000-0002-2325-413X> (IA); bormann@wiwi.uni-frankfurt.de,  <https://orcid.org/0000-0002-8131-7709> (SB); hombach@wiwi.uni-frankfurt.de,  <https://orcid.org/0000-0002-5571-3943> (KH)

Received: October 18, 2021

Revised: February 27, 2023;
December 7, 2023

Accepted: January 10, 2024

Published Online in Articles in Advance:
May 7, 2024

<https://doi.org/10.1287/mnsc.2021.03484>

Copyright: © 2024 INFORMS

Abstract. We examine the effect of trade secret protection laws on internal information integration (i.e., the extent to which economic agents are provided with access to decision-relevant information from other economic agents within a firm). We argue that stronger trade secret protection laws increase firms' internal information integration because they reduce the proprietary costs of information leakage. To test our prediction, we measure a firm's internal information integration by the share of its sites integrated into its enterprise management system. Exploiting the staggered adoption of trade secret protection laws via the Uniform Trade Secrets Act (UTSA), we find that these laws increase firms' internal information integration. This effect is stronger (weaker) for firms with higher proprietary costs (coordination benefits). Further, we provide evidence that the UTSA-induced increase in internal information integration translates into improvements of firms' internal information quality and decision-making quality. Taken together, our results enhance the understanding of the economic trade-offs shaping firms' internal information environment.

History: Accepted by Ranjani Krishnan, accounting.

Funding: S. Bormann and K. Hombach gratefully acknowledge funding from Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) [Grant Project-ID 403041268–TRR 266 Accounting for Transparency].

Supplemental Material: The online appendix and data files are available at <https://doi.org/10.1287/mnsc.2021.03484>.

Keywords: trade secret protection • UTSA • information technology • enterprise management system • information integration • internal information environment

1. Introduction

Under the knowledge-based view of the firm, a key advantage of firms over markets is their ability to create, exploit, and protect proprietary information (e.g., Liebeskind 1996, 1997). Within the boundaries of a firm, value can be generated by exchanging and using proprietary information without making it public (Robertson et al. 2015). However, firm boundaries are unlikely to perfectly protect a firm's proprietary information; corporate insiders (such as employees) constitute the most common source of information leakage, posing a major threat to a firm's competitive advantage (InfoWatch 2006, 2014, 2018; Cheng et al. 2013; Wong et al. 2019; Cabezon and Hoberg 2022). Accordingly, firms need to trade off the benefits of internally exchanging and exploiting their proprietary information against the costs associated with its increased leakage risk to outsiders.

In this paper, we study whether leakage costs affect firms' internal information integration (i.e., the extent to which firms grant their economic agents access to

decision-relevant proprietary information from other economic agents within the firm). To that end, we focus on variation in leakage costs stemming from the protection of trade secrets. Trade secrets are among firms' most valuable assets (Shapiro and Hassett 2005), and their leakage constitutes one of the biggest threats to firms' competitiveness (Wong et al. 2019).¹ To prevent such leakage from occurring, firms often rely on secrecy and restrict their economic agents' access to proprietary internal information (i.e., reduce internal information integration) (Cohen et al. 2000, Arundel 2001, Png 2017b). Another key mechanism of protecting trade secrets is trade secret protection laws (Ronde 2001). These laws enable firms to legally contest the misappropriation of trade secrets, thus reducing both the damage and the likelihood of such misappropriation by unauthorized third parties, like competitors (Franzoni and Kaushik 2016, Glaeser 2018, Li et al. 2018).

We focus on a specific trade secret protection law in the United States: the Uniform Trade Secrets Act (UTSA).

Enacted at the state level, the UTSA extends the remedies that firms can obtain in case of trade secret misappropriation and facilitates litigation, thus providing for stronger trade secret protection (Png 2017b, Glaeser 2018). We therefore argue that the UTSA reduces firms' expected cost of information leakage, causing them to increase their internal information integration (Ronde 2001, Baldwin and Henkel 2015, Elliott et al. 2019).

In our empirical tests, we measure firms' internal information integration by the share of sites² that is integrated into a firm's enterprise management system (EMS). We focus on EMSs because they are a key technology allowing firms to grant their agents access to decision-relevant information (Bloom et al. 2014). Importantly, the type of information typically integrated into EMSs—such as customer information (e.g., contact details), pricing information (e.g., product costs), or personnel information (e.g., salaries)—is covered by the increased protection of the UTSA. We focus on the share of integrated sites because the number of site managers who can contribute to and access firm-wide information increases as more sites are integrated into the firm's EMS. At the same time, as more sites are integrated into the EMS, a greater number of economic agents can potentially leak this information. Thus, we believe that the share of integrated sites is an important choice of the firm, on which the reduction in leakage costs because of trade secret protection laws has a first-order impact. We label this variable *EMS integration*.

To identify the effect of the UTSA on *EMS integration*, we employ a difference-in-differences (DID) design around states' enactments of the UTSA over the period from 1999 to 2009. During this period, the enactment of the UTSA affected sample firms in two states (i.e., Pennsylvania and Tennessee). We find that, following the enactment of the UTSA, treatment firms significantly increase their *EMS integration* relative to control firms from states with no UTSA enactment. In cross-sectional tests, we corroborate information leakage costs as the channel through which this effect arises. Specifically, we find that the UTSA increases firms' *EMS integration* more (less) when they have higher marginal costs (benefits) of internal information integration. Collectively, these findings are consistent with our prediction that trade secret protection laws increase firms' internal information integration via reducing their marginal costs of information leakage.

In further analyses, we corroborate our approach of using *EMS integration* as a measure of firms' internal information integration and document how *EMS integration* contributes to the quality of the internal information environment. Specifically, we use path analysis to document the direct and indirect effects stemming from the adoption of the UTSA on several proxies for firms' internal information quality (IIQ) and decision-making quality (DMQ), respectively. We do not find a

significant direct path from the UTSA to these proxies, consistent with idea that the UTSA affects firms' internal information environment via multiple offsetting channels. However, when we focus on the indirect path via *EMS integration*, we find a significant positive relation between the UTSA and firms' IIQ and DMQ, respectively. This result supports the notion that internal information integration, measured by *EMS integration*, is an important input to both IIQ and DMQ, via which trade secret protection laws can matter for firms' internal information environment more broadly.

In our robustness tests, we address two threats to our inferences. For one, a concern might be that our main result is confounded by omitted correlated variables. In addition to the features of our main research design that speak to this concern (such as our granular fixed effects structure, firm- and state-level time-variant controls, and event-time analyses), we provide several further pieces of evidence. First, we show that the positive relation between the UTSA and *EMS integration* holds for each of our two treatment states individually, mitigating concerns about other simultaneous events in a particular state. Second, we find that the positive effect of the UTSA is robust to different choices of control states, including a restricted set of states that have not enacted the UTSA by the end of our sample period. This finding alleviates concerns that our main result stems from fundamental differences between treatment and control firms, which may correlate with states' enactment of the UTSA before the beginning of our sample period. Third, we find that *EMS integration* is negatively associated with data breach notification laws, which increase leakage concerns as they require firms to publicly disclose data breaches. This implies that for an omitted correlated variable to confound our results, it would need to positively correlate with the UTSA and negatively correlate with data breach notification laws. Fourth, we show that our results are not driven by time-varying factors at the state and site level. Finally, our tests using the impact threshold for a correlated variable (ITCV) and robustness of inferences to replacement (RIR) suggest that it is unlikely that there is an omitted correlated variable that would render our inferences invalid.

For another, a concern might be that the UTSA affects firms' internal information integration via channels other than leakage costs (e.g., alleviated financial constraints). To address this concern, we provide two sets of tests. First, we construct an alternative dependent variable that captures firms' use of software applications that are integrated in an EMS relative to their overall use of software applications (either integrated or stand-alone). We label this variable *Application integration ratio*. Similar to our main result, we find that the UTSA increases firms' *Application integration ratios*. This finding is inconsistent with any alternative channel that

would affect firms' overall investment in software applications (either integrated or stand-alone) because such a channel would impact both the numerator and the denominator of our *Application integration ratio*. Second, we provide evidence that financial constraints or external reporting incentives are not the main channel via which the UTSA affects firms' *EMS integration* (Acharya et al. 2016).

We contribute to the literature in the following ways. First, we add to prior research on the frictions that impair information exchange between economic agents. Studies in this stream of literature emphasize that proprietary costs matter for firms' external reporting to investors (e.g., Berger and Hann 2007, Bens et al. 2011, Li et al. 2018) and for information exchange between firms (e.g., Asker and Ljungqvist 2010, Anderson et al. 2014, Christ and Nicolaou 2016, Bills et al. 2020). Whether these results generalize to the integration of information *within firms* is, however, unclear because firm boundaries could be an effective mechanism to protect proprietary information (Liebeskind 1997). Prior studies have provided analytical (Ronde 2001) and qualitative (Elliott et al. 2019) analyses on the effectiveness of firm boundaries for protecting firms' proprietary information. We provide large-scale empirical evidence on the role of leakage costs and their reduction via trade secret protection laws for firms' decisions to integrate internal information. Our study thereby complements Glaeser et al. (2023), who study where firms locate their inventors to facilitate knowledge flows in the presence of a different type of leakage concern (namely, labor market competitors).

Second, we contribute to opening the black box of what constitutes firms' internal information environment. To that end, we construct a proxy of firms' internal information integration based on their share of sites integrated into an EMS. This measure reflects an important choice variable that firms can readily adjust in response to shifts in expected costs and benefits. Thus, it provides us with a suitable setting to test theoretical predictions concerning the trade-offs that influence firms' decisions to integrate internal information (Ronde 2001). By focusing on a relevant and observable *input* to the quality of firms' internal information environment, we further complement prior research using *outcomes* related to IIQ or DMQ (e.g., Gallemore and Labro 2015, Lyle 2019).

Third, we contribute to studies on the determinants of management accounting systems (e.g., Chenhall and Morris 1986, Ittner et al. 2002, Abernethy and Bouwens 2005, Davila and Foster 2005, Amiraslani et al. 2024). Using mainly survey evidence, this literature focuses on variation in internal information demand and how it determines the design of firms' management accounting systems. Other studies have similarly focused on the implementation of EMSs and how they enrich

the internal information environment (e.g., Cassar and Gibson 2008, Heitzman and Huang 2019, Labro and Stice-Lawrence 2020, Labro et al. 2022). We complement these studies by providing large-scale, archival evidence on an important strategic cost factor, namely the costs of information leakage.

Finally, our study contributes novel evidence on the consequences of trade secret protection laws. Prior research finds that trade secret protection laws are associated with employee mobility and knowledge spillovers (Png 2017a), patenting (Png 2017b, Contigiani et al. 2018, Glaeser 2018), investment in knowledge assets (Qiu and Wang 2018), disclosure (Glaeser 2018, Li et al. 2018), and firm value (Klasa et al. 2018, Qiu and Wang 2018). More broadly, our findings contribute to the growing literature on accounting and innovation (Glaeser et al. 2023). Specifically, our results suggest an increase in internal information integration as a response to stronger protection of innovation and trade secrets.

2. Theoretical Background, Prior Literature, and Hypothesis Development

2.1. Prior Literature on Information Exchange Between Economic Agents

Our study relates to the literature on frictions that impair information exchange between economic agents. Prior studies document that proprietary costs prevent information exchange *across* firm boundaries. For instance, Christ and Nicolaou (2016) examine the use of integrated information systems in interfirm alliances. They investigate a trade-off similar to our trade-off as alliance partners need to weigh the benefits of integrated information systems (improved coordination, control, and communication within the alliance) against the costs of information exchange risk, including the proprietary costs arising from partners using the shared information opportunistically. Consistent with such trade-off, the evidence in Christ and Nicolaou (2016) suggests that firms use formal controls (e.g., policies, rules, and reporting structures) to mitigate information exchange risk. Anderson et al. (2014) discuss firms' practice of fragmenting proprietary information that is shared in alliances to mitigate the risk of information leakage. Relatedly, Clemons and Hitt (2004) discuss the concept of "poaching" (i.e., the opportunistic use of information by alliance partners) and highlight the resulting economic inefficiencies. Other studies emphasize that proprietary cost concerns prevent firms from exchanging information with specific economic agents, such as auditors (Bills et al. 2020) or banks (von Rheinbaben and Ruckes 2004, Asker and Ljungqvist 2010), that might leak such information to firms' competitors. As a consequence, firms incur higher costs in terms of audit or banking fees.

Whereas information exchange risk and resulting economic inefficiencies are prominently discussed and analyzed in the context of *interfirm* relations, it is not clear whether similar concerns apply to *intrafirm* settings. Similar to the use of contracts or formal controls in interfirm relations, firm boundaries can serve as an effective mechanism to mitigate proprietary costs of information leakage (Liebeskind 1997, Elliott et al. 2019). Yet, firm boundaries might not protect firms' proprietary information perfectly. Indeed, practitioner surveys highlight that leakage is predominantly attributable to corporate insiders (InfoWatch 2006, 2014, 2018)³ and causes major business costs (Wong et al. 2019). In line with practitioners' perceptions, Cabezon and Hoberg (2022) provide large-scale evidence that leakage of proprietary information to competitors through an important corporate insider (i.e., directors) lowers the profits and valuation outcomes of the affected firms. Finally, Glaeser et al. (2023) provide evidence suggesting that firms trade off the benefits of increased information flows against leakage costs (e.g., from potential labor market competitors) in their decisions on the location of inventors.

If firm boundaries fail to sufficiently protect firms' proprietary information, the resulting leakage concerns may affect firms' decision to integrate information internally. Consistent with this idea, Liebeskind (1997), Hannah (2005), Baldwin and Henkel (2015), and Robertson et al. (2015) provide qualitative arguments that firms restrict information provided to their managers in the presence of leakage concerns. Closely related to our research question, Ronde (2001) provides an analytical model of how expected leakage costs affect a firm's choice of organizational form (i.e., joint production or task separation). Under joint production, more employees have access to proprietary information and trade secrets. As a result, joint production is preferable to task separation in terms of operating efficiency, but also, it has higher leakage costs. These leakage costs arise because under joint production, it is more likely that employees with access to trade secrets are poached by competitors. Thus, the firm needs to trade off the benefits of better economic decisions against the costs of revealing proprietary information to competitors. Consistent with such trade-off, the author demonstrates that a firm's choice of joint production over task separation depends on the efficiency loss from task separation and the competitive environment. Similarly, Elliott et al. (2019) provide case study evidence on internal information exchange across a firm's research and development laboratories. They argue that the firm needs to trade off the benefits of improved decision making and innovation with information leakage risk. Their interview evidence suggests that relational contracts, similar to firm boundaries, importantly

mitigate such leakage risk, allowing for more information exchange.

In conclusion, practitioner surveys and analytical models suggest that firm boundaries might not always be sufficient to protect firms against information leakage. Yet, empirical evidence on the cost/benefit trade-offs shaping firms' decisions on the extent to which they provide economic agents with access to internal information is scarce. We fill this gap by providing large-scale empirical evidence on the effect of leakage costs on firms' internal information integration.

2.2. The Effect of Trade Secret Protection Laws on the Integration of Information Within Firms

When deciding on the extent to which economic agents are provided with access to information from other economic agents within the firm, firms trade off the marginal costs and benefits of internal information integration. On the one hand, access to firm-wide information improves economic agents' decision making (e.g., because they can take customer demands from other locations into account). On the other hand, the expected costs of information leakage increase as more information is made available to a greater number of economic agents (e.g., because more comprehensive, firm-wide information is more valuable to competitors and because more agents can leak such information). Thus, when deciding on their internal information integration, firms face a trade-off between the benefits of improved economic decision making and higher expected costs of information leakage.

Trade secret protection laws can increase firms' internal information integration as they reduce firms' expected costs of information leakage. For one, these laws reduce firms' costs resulting from trade secret misappropriation (Png 2017b, Glaeser 2018) as they enable firms to legally contest the disclosure and use of proprietary information by unauthorized parties. For another, trade secret protection laws reduce the probability of information leakage as they reduce third parties' benefits from access to trade secrets (Franzoni and Kaushik 2016). In line with these arguments, Ronde (2001) demonstrates analytically that firms provide more employees with access to trade secrets when trade secret protection increases.

There are also reasons why trade secret protection laws might not matter for firms' internal information integration. Firms may not need to rely on regulation to protect their trade secrets, but rather, they may use other mechanisms, such as organizational design and contracting. Moreover, firm boundaries might already sufficiently prevent information leakage (Liebeskind 1996, Elliott et al. 2019), rendering the incremental

reduction in leakage costs because of trade secret protection laws negligible.

Inconsistent with the notion that firm boundaries serve as a sufficient mechanism to protect proprietary information, however, practitioner surveys and anecdotal evidence highlight the relevance of information leakage concerns, in particular related to trade secrets (Cohen et al. 2000, National Science Foundation, National Center for Science and Engineering Statistics 2015). Thus, in line with prior analytical work (Ronde 2001), we state the following hypothesis.

Hypothesis 1. *Trade secret protection laws increase firms' internal information integration.*

3. Variable Measurement and Research Design

3.1. EMS Integration (Dependent Variable)

To test our hypothesis, we need an empirical measure that can be expected to vary with the costs and benefits of internal information integration. We obtain such a measure based on the share of sites integrated into a firm's EMS. The goal of EMSs is to automate, standardize, and integrate information across sites and functions within a firm (Ranganathan and Brown 2006, Tian and Xu 2015, Pincus et al. 2017). They provide a unified information technology (IT) infrastructure aimed at streamlining information flows, enhancing data consistency, and supporting business processes (Morris and Venkatesh 2010, Dorantes et al. 2013, Tian and Xu 2015). Importantly, as more sites are integrated into an EMS, more decision-relevant information is obtained from and distributed to more site managers within the firm.

On the benefits side, integrating more sites into an EMS improves (site) managers' decision making and related outcomes (Granlund and Malmi 2002, Ranganathan and Brown 2006, Bloom et al. 2014, Labro et al. 2022).⁴ Further, integrating more sites into a firm's EMS enables enhanced coordination among sites (Liebeskind 1997, Gallemore and Labro 2015, Labro et al. 2022).

On the costs side, firms face considerable costs of finding, implementing, customizing, and maintaining an EMS (Dorantes et al. 2013). These costs comprise (fixed) initial implementation costs as well as the incremental costs of integrating individual sites into the EMS. In addition, integrating more sites into the EMS can give rise to proprietary costs because of increased risk of information leakage (e.g., to competitors) (Ronde 2001). Next to intentional information leakage (e.g., because of theft or disclosure to unauthorized third parties) (Huth et al. 2013, InfoWatch 2018, Molok et al. 2018), users of EMSs might leak information unintentionally through unidentified channels

(Ahmad et al. 2014) or when they are uncertain about which information should not be disclosed to third parties (Molok et al. 2018). We construct our dependent variable, *EMS integration*, as the share of U.S. sites per firm, weighted by their number of employees, that have implemented an EMS by year t .

3.2. Trade Secret Protection Laws (Independent Variable)

We exploit variation in firms' exposure to trade secret protection laws introduced by the staggered enactment of the UTSA across U.S. states. Over the period from 1980 to today, the UTSA has been enacted by all states with the exception of New York. The overall goal of the UTSA is to standardize, codify, and improve state laws of trade secret protection.

The UTSA solidifies the protection of trade secrets in two respects. First, it considers the mere acquisition of a trade secret to be misappropriation, even if the information was not used or disclosed. Second, the UTSA provides for a broad definition of a trade secret, defining it as information that (1) derives independent economic value from not being generally known to or accessible by other people who may benefit from its use or disclosure and (2) is subject to reasonable efforts protecting its secrecy (e.g., via confidentiality agreements with employees). Trade secrets can be of a technical nature (software or techniques) and/or of strategic relevance (e.g., business plans, financial data, details about customers and suppliers, or personnel data). For example, the following information has been classified as a trade secret during UTSA rulings: customer identities (*Morlife Inc. v. Perry*, 56 Cal. App. 4th 1514, 1997), customer lists, preferences and contact information, pricing guidelines, historical purchasing information (*Brocade Communication Systems Inc. v. A10 Networks Inc.*, 873 F.Supp.2d 1192; N.D. Cal. 2012), internal pricing and competition strategies (*Engility Corp. v. Daniels*, No. 16-cv-2473-WJM-MEH, 2016 WL 7034976, at *6, *8; D. Colo. Dec. 2, 2016), personnel and salary information (*Marlite, Inc. v. Canas*, No. 5:09CV1401, 2009 WL 7272686, at *11; N.D. Ohio July 22, 2009), and data compilations (of public data; *AirFacts, Inc. v. de Amezaga*, 909 F.3d 84, 96-97; 4th Cir. 2018). This type of information is typically stored in EMSs or local stand-alone applications, such as accounting (e.g., financial data, internal pricing, or purchasing information), human resources (HR; e.g., compensation plans), customer relationship management (CRM; e.g., customer-related information), or supply chain management (e.g., procurement data).⁵ The definition provided by the UTSA is comprehensive as it also encompasses work in progress and negative information (e.g., information on which approaches or processes do *not* work), and it does not require the

information to be business related or to be in continuous use (Pooley 1997, Png 2017b).

To prevent actual or threatened misappropriation, the UTSA provides injunctive relief to trade secret owners. An injunction can be long enough to ensure that no advantage from misappropriation can be gained, and it allows for punitive damages up to twice the amount of the actual damages (Png 2017b).

To capture a firm's exposure to trade secret protection laws, we follow prior literature (Castellaneta et al. 2017, Glaeser 2018) and use a dummy variable indicating whether the state of a firm's headquarters has enacted the UTSA in the current year.⁶ By using a dummy variable, we refrain from subjective judgments on the strength or relative importance of the UTSA relative to pre-existing common laws (Glaeser 2018).

We use headquarters states because we expect a firm's possibilities to make claims under the UTSA to substantially increase when its headquarters state enacts the UTSA. In trade secret-related disputes, courts frequently determine personal jurisdiction based on the headquarters state because it has a substantial relationship to the case (Jones 2014, Porter 2016, Li et al. 2018). Similarly, if a firm's headquarters state has enacted the UTSA, firms can support the use of the law of the headquarters state via choice-of-law provisions (Porter 2016) and by initiating litigation in the headquarters state (Steinmeyer et al. 2019). In conclusion, we expect the enactment of the UTSA in firms' headquarters states to have an important effect on their exposure to trade secret protection and hence, on their expected leakage cost.

We confirm the importance of a firm's headquarters state for trade secret litigation by a comprehensive, empirical examination of UTSA litigation cases involving public plaintiffs (i.e., firms similar to those in our sample) during our sample period (see Online Appendix A for details). We find that in approximately 90% (80%) of all cases that fall under state (federal) jurisdiction, courts apply the UTSA of the claimant's headquarters state. This includes cases involving trade secrets misappropriation by sites or employees in states other than the headquarters state (see also Berman et al. 2017).⁷ This empirical evidence supports the use of firms' headquarters state to construct our explanatory variable.

In Table 1, panel A and Online Appendices B and C, we provide a list of UTSA enactment dates and empirical evidence on determinants of the UTSA. We do not find that enactment of the UTSA is significantly determined by state-level factors.

3.3. Data

We obtain our data from different sources. We derive site-level information on U.S. firms' EMS from the CI database provided by Aberdeen (formerly Harte

Hanks), a commercial market research firm. To collect these data, Aberdeen surveys IT managers about the type of hardware and software used at their sites. These data are sold to large IT firms (e.g., IBM, Cisco, or Dell), indicating that high timeliness and data quality as errors would be quickly detected during sales calls. The data have been widely used to study the adoption of IT (Forman 2005, Pincus et al. 2017) or implications of IT investment (Kleis et al. 2012, Bloom et al. 2014, Tian and Xu 2015, Pincus et al. 2017, Nagle 2019). For our empirical tests, we aggregate the site-level survey data to the firm level⁸ and merge the firm-level EMS data with firm-level financial information from Compustat. We provide more evidence on the CI database and our sample in Online Appendix D.

3.4. Regression Model

To test the effect of the UTSA on EMS integration, we estimate the following regression:

$$EMS\ integration_{i,t} = \beta \times UTSA_{s,t} + \theta X_{i,t} + \vartheta X_{s,t} + \partial_i + \gamma_{j,t} + \varepsilon_{i,t}, \quad (1)$$

where the subscripts i , t , s , and j are index firms, years, headquarters states, and industries, respectively. $EMS\ integration_{i,t}$ is the share of sites, weighted by the number of employees, that have been integrated into firm i 's EMS by year t . $UTSA_{s,t}$ is an indicator variable for whether headquarters state s has enacted the UTSA by year t . We provide definitions of all variables in the appendix.

Our fixed effects structure accounts for any time-invariant differences across firms via firm fixed effects (∂_i) and for industry-level shocks via industry-year fixed effects ($\gamma_{j,t}$). Because our treatment variable, $UTSA$, is defined at the state level, we cluster standard errors by state.

We include a set of firm-specific control variables, $X_{i,t}$, that have been associated with EMS implementation in prior studies (Pincus et al. 2017). Specifically, we control for firm size proxied by the natural logarithm of one plus sales ($\log(1 + Sales)$), the natural logarithm of one plus the number of employees ($\log(1 + Employees)$), and the natural logarithm of one plus the number of sites ($\log(1 + \#Sites)$) as well as for IT intensity proxied by the number of personal computers (PCs) per employee ($PCs/employee$). Further, we include an indicator variable for the use of a Big 4 audit firm (*Big 4*), average sales growth (*Avg. sales growth*), environmental uncertainty (*Uncertainty*), and unrelated firm diversification (*Unrelated diversification*).

We further include a set of time-varying, state-specific control variables, $X_{s,t}$. These variables capture contemporaneous regulatory and macroeconomic developments at

Panel A: State-level descriptives													
State	N (obs)	N (firms)	UTSA	IDD	Data breach	NCI change	State	N (obs)	N (firms)	UTSA	IDD	Data breach	NCI change
Treatment states							Control states (other; continued)						
Pennsylvania	741	99	2004	1982	2006	—	Louisiana	167	16	1981	—	2006	1992; 2002; 2004
Tennessee	317	40	2000		2005	—	Maine	19	2	1987	—	2006	—
Control states (restricted control group)							Maryland	153	25	1989	—	2008	—
Massachusetts	693	93	2018	1994	2007	—	Michigan	401	49	1998	1966	2007	—
New Jersey	612	90	2012	1987	2006	—	Minnesota	503	62	1980	1986	2006	—
New York	1,193	162	—	—	2005	—	Mississippi	45	6	1990	—	2011	2009
Texas	1,288	172	2013	1993	2009	1995	Missouri	384	48	1995	2000	2009	—
Control states (other)							Montana	20	3	1985	—	2006	—
Alabama	107	15	1987	—	2018	—	Nebraska	83	11	1988	—	2006	—
Arizona	211	31	1990	—	2006	—	Nevada	118	18	1987	—	2006	—
Arkansas	103	13	1981	1997	2005	—	New Hampshire	69	9	1990	—	2007	—
California	1,989	280	1985	—	2003	—	New Mexico	22	3	1989	—	2017	—
Colorado	243	38	1986	—	2006	1992; 2012	North Carolina	330	42	1981	1976	2005	—
Connecticut	438	54	1983	1996	2006	1992	North Dakota	11	1	1983	—	2005	—
Delaware	50	7	1982	1964	2005	—	Ohio	749	91	1994	2000	2006	1992
District of Columbia	35	4	1989	—	2007	—	Oklahoma	154	19	1986	—	2008	—
Florida	644	88	1988	1960	2014	1997	Oregon	247	33	1988	—	2007	—
Georgia	429	60	1990	1998	2005	2005	Rhode Island	61	8	1986	—	2006	—
Hawaii	19	2	1989	—	2007	2007	South Carolina	87	13	1992	—	2009	—
Idaho	51	5	1981	—	2006	1992; 2009	South Dakota	30	3	1988	—	2018	—
Illinois	737	94	1988	1989	2006	—	Utah	99	14	1989	1998	2007	—
Indiana	237	29	1982	1995	2006	—	Vermont	41	6	1996	—	2012	—
Iowa	125	15	1990	1996	2008	—	Virginia	391	54	1986	—	2008	1992; 2006
Kansas	58	12	1981	2006	2007	2008	Washington	240	29	1982	1997	2005	—
Kentucky	158	18	1990	—	2014	—	West Virginia	40	6	1986	—	2008	1984; 1992
							Wisconsin	301	34	1986	—	2006	—

Panel B: Descriptive statistics

	(1) N	(2) Mean	(3) Standard deviation	(4) p25	(5) p50	(6) p75
Main analyses (Tables 2–4)						
EMS integration	15,243	0.560	0.387	0.155	0.640	0.967
EMS integration-equally weighted	15,243	0.491	0.361	0.172	0.500	0.800
UTSA	15,243	0.726	0.446	0	1	1
log(1 + Sales)	15,243	6.717	1.785	5.447	6.722	7.942
log(1 + #Employees)	15,243	1.758	1.234	0.738	1.539	2.549
PCs/employee	15,243	0.659	0.533	0.300	0.542	0.883
log(1 + #Sites)	15,243	2.209	1.146	1.386	1.946	2.944
Uncertainty	15,243	0.060	0.114	0.012	0.024	0.054
Unrelated diversification	15,243	0.106	0.130	0	0.031	0.211
Profit	15,243	0.127	0.141	0.064	0.118	0.189
Debt ratio	15,243	0.201	0.181	0.027	0.178	0.309
Avg. sales growth	15,243	0.096	0.191	−0.008	0.072	0.165
Big 4	15,243	0.827	0.378	1	1	1
IDD	15,243	0.649	0.477	0	1	1
NCI	15,243	3.879	2.260	3	4	5
Data breach	15,243	0.297	0.457	0	0	1
GDP growth	15,243	2.268	2.702	0.677	2.351	3.917
Unemployment rate	15,243	5.311	1.556	4.425	5.092	5.933
IT employees	15,243	11.827	5.788	8.615	10.763	13.894
Competition	9,556	0.496	0.500	0	0	1
TPP	7,030	0.498	0.500	0	0	1
High-tech	15,220	0.207	0.405	0	0	0
Vertical integration	15,067	0.502	0.500	0	1	1
Path analyses (Table 5)						
Collaboration	12,434	1.150	0.285	0.952	1.129	1.315
No error restatement	13,051	−0.004	0.059	0	0	0
EA speed	12,354	−0.120	0.051	−0.148	−0.110	−0.082
Accrual quality	11,214	−0.057	0.037	−0.073	−0.048	−0.031
Capex efficiency	11,222	0.330	0.470	0	0	1
Labor efficiency	12,454	0.629	0.483	0	1	1
TFP	12,462	0.102	0.667	−0.313	0.102	0.495
Other information systems (Table 7)						
ACC-Application integration ratio	13,450	0.710	0.380	0.454	0.945	1
CRM-Application integration ratio	10,223	0.623	0.424	0.050	0.862	1
HR-Application integration ratio	11,058	0.737	0.375	0.532	0.991	1

Panel C: Pearson correlations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) EMS integration	1																
(2) UTSA	0.032*	1															
(3) log(1 + Sales)	0.082*	0.023*	1														
(4) log(1 + #Employees)	0.004	0.029*	0.859*	1													
(5) PCs/employee	0.024*	-0.054*	-0.157*	-0.202*	1												
(6) log(1 + #Sites)	-0.072*	0.012	0.652*	0.697*	-0.200*	1											
(7) Uncertainty	-0.030*	-0.036*	-0.259*	-0.223*	0.155*	-0.188*	1										
(8) Unrel. diversification	-0.011	0.007	0.176*	0.158*	-0.089*	0.293*	-0.073*	1									
(9) Profit	0.026*	-0.031*	0.292*	0.185*	-0.031*	0.138*	-0.300*	0.019*	1								
(10) Debt ratio	-0.086*	0.033*	0.194*	0.145*	-0.085*	0.202*	0.009	0.049*	0.143*	1							
(11) Avg. sales growth	-0.044*	-0.036*	0.087*	0.028*	0.049*	-0.058*	-0.128*	-0.059*	0.163*	0.061*	1						
(12) Big 4	0.091*	0.050*	0.325*	0.251*	-0.013	0.173*	-0.030*	0.063*	0.105*	0.086*	-0.002	1					
(13) IDD	0.001	-0.443*	0.049*	0.056*	-0.040*	0.119*	-0.071*	0.067*	0.011	0.021*	-0.046*	-0.046*	1				
(14) NCI	0.008	-0.015*	0.002	0.034*	-0.075*	0.072*	-0.110*	0.040*	-0.025*	0.041*	-0.026*	-0.044*	0.429*	1			
(15) Data breach	0.289*	0.112*	0.124*	0.063*	0.012	0.007	0.038*	0.017*	0.016*	-0.057*	-0.077*	0.028*	-0.093*	-0.167*	1		
(16) GDP growth	-0.162*	-0.033*	-0.101*	-0.076*	0.034*	-0.058*	0.012	-0.070*	0.023*	-0.015*	0.179*	-0.023*	-0.138*	-0.118*	-0.330*	1	
(17) Unemployment rate	0.173*	0.043*	0.089*	0.059*	0.006	0.021*	0.071*	0.038*	-0.010	-0.030*	-0.195*	0.093*	-0.050*	-0.157*	0.374*	-0.542*	1
(18) IT employees	-0.020*	-0.050*	-0.021*	-0.025*	0.072*	-0.033*	0.080*	-0.031*	-0.073*	-0.056*	0.017*	-0.010	-0.152*	-0.195*	0.073*	0.047*	-0.095*

Notes. This table provides summary statistics. Panel A provides information on the distribution of our sample firms across states along with additional state-level information. For each state, we provide the number of observations used in our main regression (Table 2), the enactment year of the UTSA, the adoption year of the IDD, the enactment year of a data breach notification law, and the year in which a change in NCI took place if any. *Treatment States* includes states affected by UTSA enactment during our sample period. *Control States (Restricted Control Group)* includes states that have not enacted the UTSA by the end of our sample period. *Control States (Other)* includes states that have enacted the UTSA prior to our sample period. Panel B provides the distributional properties of our variables, whereas panel C reports Pearson correlation coefficients. All continuous control variables, except for NCI, are winsorized at the 1st and 99th percentiles. The number of observations for each variable corresponds to the number of observations in the regression in which it is used (after dropping singletons). *EMS integration-equally weighted* is the share of sites (equally weighted) that is integrated into an EMS. Refer to the appendix for definitions of all other variables. p, percentile.

*Coefficients are significantly different from zero at the 10% level or better (two tailed).

the state level that could affect firms' information leakage risk. Specifically, we include indicator variables for whether a firm's headquarters state adopted the inevitable disclosure doctrine (IDD) (based on Klasa et al. 2018),⁹ enacted the data breach notification law (*Data Breach*) (based on Kamiya et al. 2021), and a noncompete enforceability index (NCI) (adopted from Garmaise 2011). Moreover, we include gross domestic product (GDP) growth and the level of unemployment to address the concern that an omitted economic characteristic could be correlated with both the UTSA and *EMS integration*. Finally, we also include the state-level share of employees working in the IT service sector to control for potential variation in the supply of EMSs.

Equation (1) resembles a DID specification. Our coefficient of interest is β , capturing average changes in *EMS integration* by firms located in UTSA enactment states relative to contemporaneous changes in *EMS integration* by firms located in control states. If trade secret protection laws increase firms' internal information integration, we expect β to be positive and statistically significant.

4. Results

4.1. Descriptive Statistics

Our sample starts in 1999 and ends in 2009 given the coverage of the CI database. We start our sample selection with 15,823 observations of nonfinancial U.S. firms from the CI database that we merge with Compustat based on firm name or ticker information. After dropping singletons and observations with missing values for the regression variables, we are left with 15,243 firm-year observations (corresponding to 1,916 unique firms). In our sample, 1,058 observations from two distinct states (namely, Pennsylvania and Tennessee) are exposed to an enactment of the UTSA during our sample period, corresponding to 7% of our sample (panel A of Table 1).¹⁰

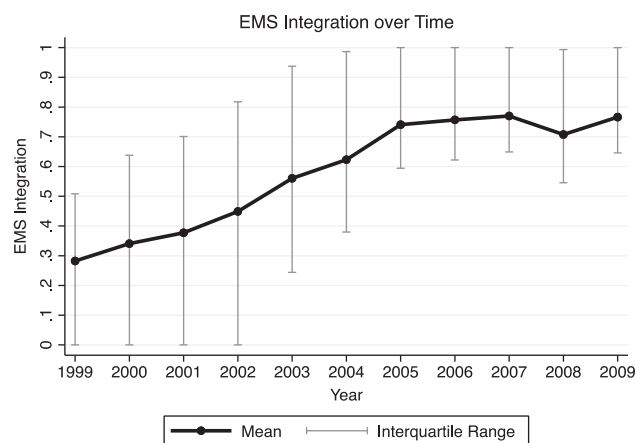
Panel B of Table 1 shows summary statistics of our main variables. Our dependent variable, *EMS integration*, is the share of sites that are integrated into a firm's EMS, weighted by its number of employees. The average firm integrates about half of its sites into an EMS. In our empirical tests, we focus on the within-firm variation in *EMS integration*, which mainly reflects firms' decision to integrate additional sites into an existing EMS rather than first-time installments of the EMS (see Online Appendix E for details). Given that larger sites are more likely to be integrated into an EMS, our employee-weighted construction of *EMS integration* is slightly higher (56%) than an equally weighted variable (i.e., the percentage share of integrated sites; 49%).¹¹ Further, the average firm in our sample has sales of about \$3,567 million, about 15,591 employees, about 19 sites, and 0.66 PCs per employee. For a comparison between

our sample and the Compustat universe, please refer to Online Appendix D. Panel C of Table 1 reports the Pearson correlation coefficients between variables. *EMS integration* correlates with several firm-level variables. In particular, larger firms and firms with higher IT intensity exhibit higher *EMS integration*.

Figure 1 shows a generally increasing time trend in firms' *EMS integration*. At the beginning of our sample period, about 30% of sites are integrated into firms' EMS on average. Until 2005, *EMS integration* roughly doubles and then stagnates at a high level of 75%. At the same time, we observe substantial cross-sectional variation in *EMS integration* as shown by overall wide yearly interquartile ranges, especially early in our sample period. Over time, interquartile ranges decrease as more firms integrate more sites into firms' EMSs, raising the lower quartile of *EMS integration*.

Panel A of Table 1 reveals that none of the UTSA-treated states adopted the IDD or experienced a change in the NCI during our sample period. Accordingly, any correlation evident in panel C of Table 1 reflects that several states have adopted both the UTSA and the IDD before our sample period. Although the UTSA-treated states adopted the data breach notification law during our sample period, these events do not coincide closely (as there is a gap of two or five years, respectively). Furthermore, the enactment of data breach notification laws likely attenuates the relation between the UTSA and firms' internal information integration as it increases the expected costs of information leakages. Overall, our descriptive statistics do not suggest that the adoption of the IDD, the adoption of the data breach notification law, or changes in the NCI are correlated with the enactment of the UTSA.

Figure 1. Development of *EMS Integration* over Time



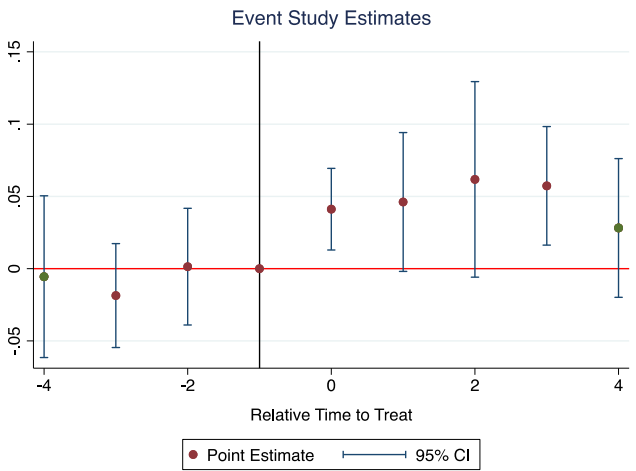
Notes. This figure plots the development of firms' *EMS integration* over time. The solid black line represents the development of yearly averages of *EMS integration*. The vertical bars show yearly interquartile ranges.

4.2. The Effect of the UTSA on EMS Integration

In Figure 2, we present event-time plots illustrating the effect of trade secret protection laws on firms’ internal information integration following Barrios (2022). The figure shows lead-lag coefficients from a two-way fixed effects regression (including firm and industry-year fixed effects but no linear controls) of *UTSA* on *EMS integration*. The *x* axis presents the distance in years from the year in which a firm’s headquarters state adopted the UTSA. We use the year prior to the UTSA enactment as a baseline. The figure shows a strong increase (relative to the baseline year) in coefficients around UTSA enactments (i.e., from event-year $t = 0$ onward). Further, the coefficients stay at elevated levels during the three years following the UTSA enactment. Thus, the figure provides prima facie evidence of a positive and persistent effect of *UTSA* on *EMS integration*.¹²

We complement this evidence with the DID regression results reported in Table 2. Column (1) in Table 2 includes only firm and industry-year fixed effects as control variables to alleviate the concern that including covariates affected by the treatment can impair the ability to draw causal inferences (Gormley and Matsa 2014, Jennings et al. 2023). Column (2) in Table 2 further adds controls for time-varying firm characteristics, whereas column (3) in Table 2 adds state-level time-varying controls. The coefficients on *UTSA*¹³ are of similar magnitude and statistically significant at the 1% level across all specifications.¹⁴ This finding supports our prediction

Figure 2. (Color online) Event Study Estimates



Notes. This figure plots event-study estimates from a two-way fixed effect regression (including firm and industry-year fixed effects but no linear controls) of the effect of the UTSA on firms’ *EMS integration*. We use the year prior to the UTSA enactment as the baseline. The gray symbols present point estimates of the lead/lag coefficients. The vertical bars indicate 95% confidence intervals (95% CIs). Standard errors are clustered at the state level.

Table 2. Effect of Trade Secret Protection Laws on Firms’ Internal Information Integration

	(1) EMS integration	(2) EMS integration	(3) EMS integration
UTSA	0.044*** (2.92)	0.046*** (3.01)	0.045*** (2.80)
log(1 + Sales)		0.024 (1.48)	0.023 (1.42)
log(1 + #Employees)		−0.012 (−0.40)	−0.011 (−0.37)
PCs/employee		0.031*** (3.43)	0.031*** (3.43)
log(1 + #Sites)		−0.023*** (−2.72)	−0.024*** (−2.70)
Uncertainty		0.059 (1.02)	0.060 (1.05)
Unrelated diversification		−0.160*** (−3.82)	−0.160*** (−3.80)
Profit		−0.016 (−0.30)	−0.017 (−0.31)
Debt ratio		−0.010 (−0.30)	−0.009 (−0.29)
Avg. sales growth		−0.028 (−1.11)	−0.028 (−1.12)
BIG 4		0.006 (0.37)	0.005 (0.36)
IDD			−0.011 (−0.55)
NCI			0.006 (0.89)
Data Breach			−0.017** (−2.53)
GDP growth			0.001 (0.61)
Unemployment rate			−0.005 (−1.00)
IT employees			−0.000 (−0.05)
Observations	15,243	15,243	15,243
Firm FE	Yes	Yes	Yes
Industry × year FE	Yes	Yes	Yes
Within R ²	0.001	0.009	0.010
Adjusted R ²	0.680	0.683	0.683

Notes. This table reports estimates from regressing the UTSA on firms’ *EMS integration*. The dependent variable, *EMS integration*, is the share of sites (weighted by their number of employees) integrated into a firm’s EMS by year t . The main independent variable is *UTSA*, an indicator for whether a firm’s headquarters state has enacted the UTSA by year t . Models are estimated using a pooled ordinary least squares (OLS) regression specification over the period 1999–2009. All continuous control variables, except for *NCI*, are winsorized at the 1st and 99th percentiles. We report t -statistics based on standard errors clustered at the state level in parentheses. Refer to the appendix for variable definitions. FE, fixed effect.

Two-tailed significance levels at the 5% level; *two-tailed significance levels at the 1% level.

that the enactment of the UTSA reduces firms' expected costs of information leakage, thus increasing their internal information integration. In terms of economic significance, the 0.045 coefficient in column (3) in Table 2 translates into a 23% increase of a standard deviation in *EMS integration* because of the UTSA.¹⁵

To gauge the validity of the parallel trends assumption underlying our DID approach, we introduce lead-lag terms for the four periods before and after UTSA enactments in our main specification. Similar to Figure 2, this test provides an event-time analysis of the effect of the UTSA but adds linear controls. $UTSA^{-4}$, $UTSA^{-3}$, and $UTSA^{-2}$ take the value of one if a firm's headquarters state is going to enact the UTSA in four, three, or two years, respectively, and zero otherwise. These coefficients capture differences in pre-UTSA trends between treatment and control firms relative to the baseline year (i.e., one year before the UTSA enactment). $UTSA^0$, $UTSA^{+1}$, $UTSA^{+2}$, $UTSA^{+3}$, and $UTSA^{+4+}$ takes the value of one if a firm's headquarters state enacted the UTSA in the current year or one, two, three, or four or more years ago, respectively. These coefficients capture differences between treated and control firms after UTSA enactments relative to the baseline year.

The parallel trends assumption is supported as long as states' enactments of the UTSA are otherwise exogenous with respect to changes in *EMS integration*, as documented by Ribstein and Kobayashi (1996). Estimates in Table 3 show that the preperiod coefficients on $UTSA^{-2}$, $UTSA^{-3}$, and $UTSA^{-4}$ are individually and jointly insignificant ($p = 0.692$), which does not support the notion of a differential pre-UTSA trend in *EMS integration* between treatment and control firms. By contrast, the coefficients relating to the post-UTSA period are individually and jointly significant ($p = 0.026$), indicating a positive and persistent effect of the UTSA on *EMS integration*.

4.3. Information Leakage Costs Channel

In this section, we provide empirical evidence substantiating the channel through which we predict that trade secret protection laws affect firms' internal information integration: the costs of information leakage. To this end, we differentiate between firms whose internal information integration is primarily determined by marginal costs versus marginal benefits. If trade secret protection laws affect firms' internal information integration via leakage costs (as we predict), we expect firms for which marginal costs (benefits) matter more to increase their internal information integration more (less) in response to the UTSA. This follows as a reduction in leakage costs should have a stronger (weaker) impact for firms with higher marginal costs (benefits) of internal information integration.

To explore the differential effects of the UTSA with respect to firms' marginal costs/benefits of internal

Table 3. Dynamic Effect of Trade Secret Protection Laws on Firms' Internal Information Integration

	(1) <i>EMS integration</i>
$UTSA^{-4}$	−0.006 (−0.22)
$UTSA^{-3}$	−0.017 (−0.88)
$UTSA^{-2}$	−0.003 (−0.13)
$UTSA^0$	0.038** (2.38)
$UTSA^{+1}$	0.042 (1.57)
$UTSA^{+2}$	0.063** (2.17)
$UTSA^{+3}$	0.058*** (2.87)
$UTSA^{+4+}$	0.030 (1.15)
Time-varying firm and state controls	Yes
Observations	15,243
Firm FE	Yes
Industry × year FE	Yes
Within R^2	0.010
Adjusted R^2	0.683
F: $UTSA^{-4} + UTSA^{-3} + UTSA^{-2} = 0$	0.692
F: $UTSA^0 + UTSA^{+1} + UTSA^{+2} + UTSA^{+3} + UTSA^{+4+} = 0$	0.026

Notes. This table reports estimates of the dynamic effect of the UTSA on firms' *EMS integration*. The dependent variable, *EMS integration*, is the share of sites (weighted by their number of employees) integrated into a firm's EMS by year t . $UTSA^{-4}$, $UTSA^{-3}$, and $UTSA^{-2}$ are equal to one if the firm is headquartered in a state that will enact the UTSA in four, three, or two years, respectively; $UTSA^0$, $UTSA^{+1}$, $UTSA^{+2}$, $UTSA^{+3}$, and $UTSA^{+4+}$ are equal to one if the firm is headquartered in a state that enacted the UTSA in the current year or enacted the UTSA one, two, three, or four and more years ago, respectively and zero otherwise. The same set of control variables as in Table 2 is included but not reported for conciseness. Models are estimated using a pooled OLS regression specification over the period 1999–2009. All continuous control variables, except for *NCL*, are winsorized at the 1st and 99th percentiles. We report t -statistics based on standard errors clustered at the state level in parentheses. Refer to the appendix for variable definitions. FE, fixed effect.

Two-tailed significance levels at the 5% level; *two-tailed significance levels at the 1% level.

information integration, we estimate the following cross-sectional regressions:

$$\begin{aligned}
 &EMS\ integration_{i,t} \\
 &= \beta_1 \times CROSS_{i,t} + \beta_2 \times UTSA_{s,t} \times CROSS_{i,t} \\
 &\quad + \theta X_{i,t} + \partial_i + \eta_{s,t} + \gamma_{j,t} + \varepsilon_{i,t},
 \end{aligned} \tag{2}$$

where $EMS\ integration_{i,t}$, $UTSA_{s,t}$, $X_{i,t}$, ∂_i , and $\gamma_{j,t}$ are defined. $CROSS_{i,t}$ is one of the cross-sectional variables described reflecting differences in firms' marginal costs or marginal benefits of internal information integration.

$\eta_{s,t}$ denotes state-year fixed effects, which subsume the state-level time-varying controls from Equation (1). That is, in our cross-sectional tests, we compare the UTSA effect on *EMS integration* depending on differences in firms' marginal costs or marginal benefits of internal information integration while removing firm and time-varying state and industry effects. We cluster standard errors by state.

To capture differences in firms' *marginal costs* of internal information integration, we use two measures of external competition. Trade secrets are likely more important for firms facing stronger competition from existing rivals (Eisenhardt and Martin 2000, Li and Li 2020). Accordingly, we expect firms to incur greater marginal costs from internal information integration and thus, to respond more strongly to the UTSA when they face greater external competitive pressure. Our first measure of external competition is based on management's discussion of competition in their 10-K filings (*Competition*) following Li et al. (2013). Specifically, the measure is the ratio between the number of references to competition and the total number of words in a 10-K filing. Our second measure is firms' technological peer pressure (*TPP*) following Cao et al. (2018), which more specifically captures firms' exposure to

technological competition. Moreover, we expect firms to exhibit higher marginal costs of internal information integration when they belong to a high-tech industry. These firms typically have more valuable inside information, making information leakage particularly costly (Aboody and Lev 2000). Following Francis and Schipper (1999), *High Tech* indicates whether a firm belongs to a high-tech industry.

Firms with greater organizational complexity typically exhibit larger *marginal benefits* of internal information integration. This follows as more complex firms face a greater need for internal coordination (Godigbe et al. 2022), and internal information integration can improve coordination within a firm (Labro et al. 2022). We proxy for organizational complexity using vertical integration as measured by Frésard et al. (2020). Vertical integration reflects the potential of a firm's products to be vertically integrated to other products offered by the same firm. Firms with greater vertical integration will likely decide on their internal information integration primarily because of their (larger) marginal benefits. Accordingly, we expect them to respond *less* to the shift in marginal costs because of the UTSA.¹⁶

Table 4 reports our cross-sectional results. Columns (1) and (2) in Table 4 show that the coefficients on

Table 4. Differential Effects of Trade Secret Protection Laws on Firms' Internal Information Integration

	Proprietary costs			Coordination benefits
	(1) <i>EMS integration</i>	(2) <i>EMS integration</i>	(3) <i>EMS integration</i>	(4) <i>EMS integration</i>
<i>UTSA × Competition</i>	0.042** (2.44)			
<i>UTSA × TPP</i>		0.102* (1.98)		
<i>UTSA × High-tech</i>			0.096* (1.78)	
<i>UTSA × Vertical integration</i>				−0.038** (−2.06)
Time-varying firm controls	Yes	Yes	Yes	Yes
Observations	9,556	7,030	15,220	15,067
Firm FE	Yes	Yes	Yes	Yes
Industry × year FE	Yes	Yes	Yes	Yes
State × year FE	Yes	Yes	Yes	Yes
Within R ²	0.009	0.010	0.010	0.010
Adjusted R ²	0.672	0.673	0.686	0.687

Notes. This table reports estimates of the differential effects of the UTSA enactment on firms' *EMS integration* for firms facing higher proprietary costs (*Competition*, *TPP*, and *High-tech* industry) or firms with larger coordination needs (*Vertical integration*). The dependent variable, *EMS integration*, is the share of sites (weighted by their number of employees) integrated into a firm's EMS by year t . The main independent variables are the interaction terms between *UTSA* and the respective cross-sectional variable. The same set of time-varying firm control variables as in Table 2 as well as the respective cross-sectional variables (on a stand-alone basis; i.e., not interacted) are included but not reported for conciseness. The coefficients on *UTSA* and state-level controls are absorbed by the state × year fixed effect (FE). Models are estimated using a pooled OLS regression specification over the period 1999–2009. All continuous control variables, except for *NCI*, are winsorized at the 1st and 99th percentiles. We report t -statistics based on standard errors clustered at the state level in parentheses. Refer to the appendix for variable definitions.

*Two-tailed significance levels at the 10% level; **two-tailed significance levels at the 5% level.

the interaction terms $UTSA \times Competition$ and $UTSA \times TPP$ are positive and statistically significant at the 5% level and 10% level, respectively, indicating a stronger effect of the UTSA for firms facing more competition. Similarly, the coefficient on the interaction term $UTSA \times High\text{-}tech$ (column (3) in Table 4) is positive and statistically significant at the 10% level, indicating a stronger effect of the UTSA for firms belonging to the high-tech industry. The coefficient on the interaction term $UTSA \times Vertical\ Integration$ (column (4) in Table 4) is, on the other hand, negative and statistically significant at the 5% level, indicating that the effect of the UTSA is weaker for firms with greater vertical integration. Overall, these cross-sectional results are consistent with information leakage costs as the channel through which trade secret protection laws affect firms' internal information integration.

4.4. The Effect of the UTSA on Firms' Internal Information Environment via EMS Integration

In our main tests, we focus on the effect of the UTSA on *EMS integration*. Via its effect on *EMS integration*, the UTSA may also enrich the quality of firms' internal information environment. This follows as internal information integration is an important input to firms' IIQ and DMQ (e.g., Bouwens and Abernethy 2000). However, the UTSA may also affect IIQ and DMQ via other channels than *EMS integration*. For instance, it could reduce information spillovers from local peer firms (Hellmann and Perotti 2011, Li and Sandino 2018), which could offset a positive information effect of greater *EMS integration* and ultimately, even decrease IIQ and DMQ. Hence, the overall relation between UTSA and these outcomes of firms' internal information environment is a priori unclear.

In this section, we investigate the direct and indirect (via *EMS integration*) paths from UTSA to firms' internal information environment captured by several proxies of firms' IIQ and DMQ, respectively. For IIQ, we use four proxies: (1) collaboration culture (Chen et al. 2022), (2) earnings announcement speed (Brazel and Dang 2008, Gallemore and Labro 2015, Godigbe et al. 2022), (3) absence of restatements because of unintentional errors (Hennes et al. 2008, Gallemore and Labro 2015, Chen et al. 2018), and (4) accrual quality (Dechow and Dichev 2002, Morris and Venkatesh 2010, Labro et al. 2022). We expect firms with greater *EMS integration* to exhibit (1) a more collaborative corporate culture because they have the necessary technology to facilitate information exchange, enabling a strong communicative and collaborative environment. We further expect firms with higher *EMS integration* to have (2) higher earnings announcement speed, (3) a lower likelihood of restatements because of unintentional errors, and (4) a higher accrual quality because of better integrated and streamlined reporting processes, enabling them to

produce high-quality information more quickly (Chen et al. 2018, Labro et al. 2022).

For DMQ, we use three proxies: (1) capex efficiency (Goodman et al. 2014), (2) labor efficiency (Pinnuck and Lillis 2007, Jung et al. 2014), and (3) total factor productivity (Imrohoroglu and Tuzel 2014). If firms with higher *EMS integration* have better information for decision making, we expect them to allocate their resources more efficiently, as reflected by higher values of these three variables.

We perform path analyses and specify the following structural equation models of the UTSA's direct and indirect relations (via *EMS integration*) to IIQ and DMQ outcomes, respectively:

$$EMS\ integration_{i,t} = \beta \times UTSA_{s,t} + \theta X_{i,t} + \vartheta X_{s,t} + \partial_i + \gamma_{j,t} + \varepsilon_{i,t} \quad (3)$$

$$Y_{i,t} = \alpha \times UTSA_{s,t} + \delta \times EMS\ integration_{i,t} + \rho X_{i,t} + \tau X_{s,t} + \partial_i + \gamma_{j,t} + \mu_{i,t} \quad (4)$$

Equation (3) resembles our main specification, where β reflects the direct path from UTSA to *EMS integration*. Equation (4) regresses both UTSA and *EMS integration* on the IIQ and DMQ outcomes defined, where α (δ) captures the direct path between UTSA (*EMS integration*) and these outcomes. Finally, $\delta \times \beta$ captures the indirect path (via *EMS integration*) from UTSA to these outcomes. In both equations, we use the same controls and fixed effects as in our main test.¹⁷ Following Zhao et al. (2010), we bootstrap standard errors. If the UTSA affects IIQ and DMQ, respectively, via *EMS integration*, we expect the products of β (the direct path from UTSA to *EMS integration*) and δ (the direct path from *EMS integration* to our outcome variables) to be positive and significant ($\beta \times \delta > 0$).

Table 5 shows the results of our path analyses. As in our main result, we document a significantly positive relation between UTSA and *EMS integration* across all specifications, consistent with a positive impact of trade secret protection laws on firms' internal information integration. We, however, fail to document a significant direct path from UTSA to almost all proxies of IIQ or DMQ, with the only exception of a positive relation between UTSA and earnings announcement speed (two-tailed $p=0.08$). These findings suggest that the UTSA has no consistent *direct* effect on the internal information environment. At the same time, we consistently observe a positive direct path from *EMS integration* to almost all IIQ and DMQ outcomes (except for capex efficiency, which is marginally insignificant with a two-tailed p -value of 0.12). This finding is consistent with the notion that *EMS integration* is an important input factor to the quality of firms' internal information environment. Similarly, we document a positive and

Table 5. Effect of Trade Secret Protection Laws via Information Integration on Firms’ Internal Information Environment

	IIQ				DMQ		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>Collaboration</i>	<i>No error restatement</i>	<i>EA speed</i>	<i>Accrual quality</i>	<i>Capex efficiency</i>	<i>Labor efficiency</i>	<i>TFP</i>
Direct path							
$P(UTSA, Outcome) (\alpha)$	−0.046 (−0.86)	0.003 (0.56)	0.064* (1.77)	−0.005 (−0.08)	−0.009 (−0.33)	−0.013 (−0.39)	−0.002 (−0.10)
Indirect path							
$P(UTSA, EMS) (\beta)$	0.148*** (3.61)	0.132*** (3.33)	0.108*** (2.61)	0.137** (2.28)	0.167*** (3.92)	0.151*** (3.78)	0.136*** (3.39)
$P(EMS, Outcome) (\delta)$	0.024* (1.88)	0.002*** (2.66)	0.015* (1.79)	0.026** (2.54)	0.010 (1.58)	0.017** (2.21)	0.015*** (2.99)
$P(UTSA, EMS) \times P(EMS, Outcome) (\beta \times \delta)$	0.003* (1.70)	0.0003** (2.10)	0.002 (1.43)	0.004* (1.71)	0.002 (1.47)	0.002* (1.89)	0.002** (2.27)
Total effect							
$P(UTSA, Outcome) + P(UTSA, EMS) \times P(EMS, Outcome)$	−0.043 (−0.80)	0.003 (0.62)	0.065* (1.81)	−0.002 (−0.03)	−0.007 (−0.26)	−0.010 (−0.31)	0.001 (0.02)
Time-varying firm and state controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,434	13,051	12,354	11,214	11,222	12,454	12,462
Firm FE	Yes	Yes	Yes	No	Yes	Yes	Yes
State FE	No	No	No	Yes	No	No	No
Industry $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. This table reports coefficients from path analyses. We obtain coefficient estimates from the following system of equations:

$$EMS\ integration_{i,t} = \beta \times UTSA_{s,t} + \theta X_{i,t} + \vartheta X_{s,t} + \partial_i + \gamma_{j,t} + \varepsilon_{i,t}$$
$$Y_{i,t} = \alpha \times UTSA_{s,t} + \delta \times EMS\ integration_{i,t} + \rho X_{i,t} + \tau X_{s,t} + \partial_i + \gamma_{j,t} + \mu_{i,t},$$

where Y_{it} is one of the four (three) outcome variables capturing firms’ IIQ (DMQ) reported in columns (1)–(4) (columns (5)–(7)). The same set of control variables as in Table 2 is included but not reported for conciseness. In columns (5) and (6), the covariates from the first-stage regressions are also included as controls. All continuous variables are standardized and except for *NCI*, winsorized at the 1st and 99th percentiles. We report *t*-statistics based on bootstrapped standard errors. Refer to the appendix for variable definitions. We further provide Pearson correlation coefficients between *EMS integration* and our proxies for internal information environment in Online Appendix F. FE, fixed effect.

*Two-tailed significance levels at the 10% level; **two-tailed significance levels at the 5% level; ***two-tailed significance levels at the 1% level.

significant indirect path ($\delta \times \beta$), via *EMS integration*, from *UTSA* to almost all of our outcomes, except for earnings announcement speed (two-tailed $p = 0.15$) and capex efficiency (two-tailed $p = 0.14$).

In conclusion, the results of our path analyses highlight that the overall effect of the *UTSA* on firms’ internal information environment is ambiguous, consistent with its nuanced effects on firms’ external information environment documented in Glaeser (2018). By contrast, we find evidence for a positive relation between *EMS integration* and IIQ/DMQ as well as a positive indirect relation between the *UTSA* and IIQ/DMQ via *EMS integration*. This finding supports the notion that the *UTSA*-induced increases in *EMS integration* matter because they translate into higher IIQ and DMQ.

5. Robustness Tests

In this section, we probe the robustness of our results with regard to two key concerns: first, that the relation between *UTSA* and *EMS integration* is confounded by omitted correlated variables and second, that the *UTSA* affects firms’ information integration via channels other than leakage costs.

5.1. Omitted Correlated Variables

In our main analyses, we include a granular fixed effects structure, time-varying control variables, and event-time analyses documenting a common trend between treatment and control states prior to *UTSA* enactments. This approach restricts confounding factors to time-varying firm-level or state-level factors that change around states’ enactments of the *UTSA* and are not accounted for by our control variables. In this section, we further probe the robustness of our main finding to omitted variables and potential confounding events.

5.1.1. Treatment Effects by Individual State. In our main tests, we pool the effect of the *UTSA* on *EMS integration* across all firms headquartered in states that enact the law during our sample period. Because our treatment effect originates from only two states, there is a concern that contemporaneous events occurring in either one of these states may confound our results. To address this concern, we document that the effect of the *UTSA* holds for each treatment state individually (Table 6, panel A).¹⁸ If our inferences were affected by a confounding event, such an event would need to take place in each of our treatment states at different

Table 6. Confounding Events and Omitted Correlated Variables

Panel A: Treatment effects separately by state				
	(1) <i>EMS integration</i> Pennsylvania	(2) <i>EMS integration</i> Tennessee		
<i>UTSA</i>	0.052*** (4.66)	0.105*** (6.57)		
Time-varying firm and state controls	Yes	Yes		
Observations	14,499	14,064		
Firm FE	Yes	Yes		
Industry $\times$ year FE	Yes	Yes		
Within R^2	0.012	0.012		
Adjusted R^2	0.684	0.682		
Panel B: Restricted control states				
	(1) <i>EMS integration</i> UTSA	(2) <i>EMS integration</i> IDD	(3) <i>EMS integration</i> NCI	(4) <i>EMS integration</i> data breach
<i>UTSA</i>	0.089*** (9.39)	0.050*** (3.06)	0.048*** (2.74)	0.113* (2.92)
<i>Time-varying firm and state controls</i>	Yes	Yes	Yes	Yes
Observations	4,740	14,046	14,082	1,983
Firm FE	Yes	Yes	Yes	Yes
Industry $\times$ year FE	Yes	Yes	Yes	Yes
Within R^2	0.019	0.011	0.008	0.050
Adjusted R^2	0.683	0.680	0.682	0.687
Panel C: Site-level characteristics				
	(1) <i>EMS adoption</i> _{site} site-level regression	(2) <i>EMS integration</i> _{FE}		
<i>UTSA</i>		0.112** (2.01)		
$\log(1 + \#Employees)_{site}$		0.200*** (25.50)		
<i>PCs/employee</i> _{site}		0.006*** (3.58)		
Time-varying firm and state controls		No		
Observations		151,286		
Firm FE		Yes		
Industry $\times$ year FE		No		
Site state		Yes		
Within R^2		No		
Pseudo-(adjusted) R^2		0.033		
		0.061		
		0.588		

Notes. This table reports robustness tests relating to potential confounding factors. In panel A, we provide estimates of the effect of the UTSA separately for each of our two treatment states. In column (1) (column (2)), firm-level changes in *UTSA* only originate from the enactment of the UTSA in Pennsylvania (Tennessee). Panel B reports estimates of the main effect of *UTSA* on *EMS integration* using different control groups. In column (1), the control group only includes firms from states that have not enacted the UTSA by 1999 (when our sample period starts). Specifically, the control group contains firms headquartered in Massachusetts, New Jersey, New York, or Texas. In column (2), the control group excludes firms that are headquartered in states adopting the IDD during our sample period (i.e., Kansas, Missouri, and Ohio). In column (3), the control group excludes firms that are headquartered in states exhibiting a change in their noncompete enforcement index during our sample period (i.e., firms from Georgia, Hawaii, Idaho, Kansas, Louisiana, Mississippi, and Virginia). In column (4), the control group excludes firms that are headquartered in a state that enacted a data breach notification law during our sample period (i.e., firms from Arizona, Arkansas, California, Colorado, Connecticut, Delaware, District of Columbia, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Louisiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Missouri, Montana, Nebraska, Nevada, New Hampshire, New Jersey, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Puerto Rico, Rhode Island, South Carolina, Texas, Utah, Virgin Islands, Virginia, Washington, West Virginia, and Wisconsin). In panel C, we re-estimate our main regression controlling for site-level characteristics following Pierri and Timmer (2020); column (1) reports determinants of *EMS adoption* at the site level. In column (2), the firm fixed effect (FE) from a year-by-year estimation of the site-level regression in column (1) (*EMS integration*_{FE}) is used as an alternative dependent variable. In column (2) in panels A–C, the same set of control variables as in Table 2 is included but not reported for conciseness. Models are estimated using a pooled OLS regression specification over the period 1999–2009. All continuous control variables, except for *NCI*, are winsorized at the 1st and 99th percentiles. We report *t*-statistics based on standard errors clustered at the state level in parentheses. In panel B, columns (1) and (4), we assign significance levels based on bootstrapped confidence intervals because of the small number of clusters (Cameron et al. 2008). Refer to the appendix for variable definitions.

*Two-tailed significance levels at the 10% level; **two-tailed significance levels at the 5% level; ***two-tailed significance levels at the 1% level.

points in time, coinciding with each state's enactment of the UTSA (Wang and Burke 2022).

5.1.2. Sensitivity to Choices of Control States. By the start of our sample period, most states have already enacted the UTSA. Thus, one might be concerned that there are fundamental differences between our treatment states (that enact the UTSA during our sample period) and those control states that had already enacted the UTSA by the beginning of our sample period. Similarly, our control states are exposed to other regulatory events during our sample period (Table 1, panel A), which could also affect our inferences.

To address these concerns, we follow Ham and Ueda (2021) and re-estimate Equation (1) using different sets of control states. First, we restrict our set of control states to those that had not enacted the UTSA by 1999 (the beginning of our sample period).¹⁹ Second, we drop states from the control group that experience other regulatory events during our sample period. Specifically, we exclude firms from control states that during our sample period, adopt the IDD, exhibit a change in their noncompete enforcement index, or enact a data breach notification law. Our findings in columns (2)–(4) in panel B of Table 6 remain unchanged. Similarly, we find that the parallel trend assumption holds across all four sets of control states (untabulated). Collectively, these findings refute the notion that our inferences are affected by differences between treatment and control firms.

5.1.3. Data Breach Notification Laws. Next, we corroborate our main finding with an alternative event that we expect, like the enactment of the UTSA, to affect firms' information leakage costs: the enactment of data breach disclosure notification laws. These laws require firms to notify affected parties (e.g., consumers) of a data breach after its discovery (Romanosky et al. 2011, Chen et al. 2019), thereby imposing costs on firms in the form of reputation concerns (Romanosky et al. 2011), increased stakeholder concerns about data cyber risk management (Chen et al. 2019), and investments in information security (Romanosky et al. 2011). That is, data breach notification laws increase the costs of information leakages resulting from data breaches, which in turn, may incentivize firms to reduce *EMS integration*. Thus, compared with the UTSA, we expect data breach notification laws to affect *EMS integration* in the opposite direction.

In column (3) of Table 2, the coefficient on *Data breach* captures the effect of data breach notification laws on *EMS integration* (see Kamiya et al. 2021 and Table 1, panel A for an overview of the (relevant) effective enactment dates). We find a significant negative relation between *Data breach* and *EMS integration*. This finding mitigates concerns that our main result is driven by

unobservable factors correlated with states' enactment of the UTSA because such factors would also need to correlate in the opposite direction with the enactment of data breach notification laws. It further refutes the notion that *EMS integration* reflects firms' decision to enhance information security rather than information integration, in line with the evidence in Bloom et al. (2014).²⁰

5.1.4. State-Level UTSA Determinants. Following Kim et al. (2021), we further explore the concern that state-level correlated omitted variables are driving our results. We do this by regressing *UTSA* on *Avg. EMS integration*, which is the average of *EMS integration* across all firms in a given state over the past three years, and several other state-level characteristics for the same three-year average. The results presented in Online Appendix C indicate that neither *Avg. EMS integration* nor other state-level characteristics have any predictive power for the UTSA. In line with this, Ribstein and Kobayashi (1996) present evidence that the adoption of the UTSA is not driven by lobbying interests but rather, is associated with efficiency benefits of uniformity and the effort of the Commissioners on Uniform State Laws (Glaeser 2018). Overall, these results lend support to the notion that UTSA enactment is plausibly exogenous and does not stem from pre-existing state-level characteristics or *EMS integration*.

5.1.5. Site-Level Factors. We perform our tests at the firm level because first, the applicability of the UTSA is strongly determined by a firm's headquarters state (see Section 3.2) and second, decisions to integrate sites into an EMS are typically made by the firm's top management rather than individual sites (Cheng et al. 2021). Yet, one might be concerned that the integration of a site into an EMS is also importantly determined by site-level characteristics, such as its size or IT sophistication, which are not appropriately captured by our firm-level average control variables and add noise to our empirical tests.

To assess the sensitivity of our results with respect to site-level characteristics, we adopt the approach of Pierri and Timmer (2020) and measure firms' *EMS integration* controlling for site-level aspects that are potential confounders of firms' decision to integrate internal information. Specifically, we construct an alternative measure of *EMS integration* based on year-by-year site-level regressions of a site's integration into an EMS on a firm fixed effect while controlling for several site-level characteristics. Specifically, our controls include the size of the site (proxied by the natural logarithm of one plus the number of employees at the site level), its IT sophistication (proxied by the number of personal computers divided by the number employees at the site level), and location (i.e., site-state fixed effects

accounting for, for example, differences in local taxes as well as labor and product markets). Our alternative measure, $EMS\ integration_{FE}$, is provided by the firm fixed effects from these regressions. Notably, this measure is purged of the variation arising from the aforementioned site-level characteristics (see also Beaudry et al. 2010 for a similar approach measuring IT adoption at a regional level). Table 6, panel C displays our results. As expected, we find that larger sites and sites with a higher degree of IT sophistication are significantly more likely to be integrated into a firm's EMS (column (1) in panel C of Table 6). Notably, accounting for these characteristics does not change our inferences with respect to the effect of the UTSA on $EMS\ integration$ (column (2) in panel C of Table 6). This finding does not support a major confounding effect of site-level characteristics on our main finding.

5.1.6. Impact of Potential Omitted Correlated Variables. Finally, we quantify the impact of potential omitted correlated variables following Frank (2000), Larcker and Rusticus (2010), and Busenbark et al. (2022). Specifically, we compute the ITCV to determine the minimum impact required for an omitted correlated variable to invalidate our inferences. The impact of a variable is defined as the product of its partial correlations (after the other control variables and fixed effects) with $UTSA$ and $EMS\ integration$, respectively. In untabulated tests, we find that the ITCV for $UTSA$ is 0.0116 at the 10% significance level. This means that the impact of an omitted correlated variable would need to exceed 0.0116 to invalidate our inferences. To put this threshold into perspective, we examine the impact scores of the control variables from our main model. Among these variables, IDD exhibits the highest impact score with 0.0021. Notably, the threshold value exceeds this score by more than 5.5 times.

Similarly, the RIR analysis suggests that 46% of the parameter estimate of $UTSA$ would have to be because of bias to invalidate our inferences at the 10% level. Assuming that we include a reasonable set of control variables in our model, both tests suggest that it is unlikely that an omitted correlated variable exists that would render our main results insignificant if included in our main model (Busenbark et al. 2022, Drake et al. 2022).

5.2. Alternative Channels

Another concern is that the UTSA might affect firms' internal information integration via channels other than leakage cost. To address this concern, we first present analyses corroborating the leakage costs channel by using an alternative dependent variable, which captures firms' internal information integration after accounting for other investments into information system. Second, we probe the sensitivity of our results to

two specific alternative channels, namely firms' external reporting incentives and financial constraints.

5.2.1. Other Investments in Information System. First, we address the concern that the UTSA might incentivize managers to invest in firms' overall information systems (not only in those integrated into an EMS) for reasons other than reduced leakage costs (such as alleviated financial constraints or increased incentives to digitalize information). To that end, we leverage our granular data on specific software applications. Specifically, we examine the effect of the UTSA on firms' use of three specific software applications: ACC, HR, and CRM.²¹ These applications can be part of an EMS or installed as a stand-alone application. A key difference between integrated (EMS) and stand-alone applications is that EMSs enable information *integration* across sites and functions, whereas stand-alone applications only allow for the exchange of information across sites within a given function (e.g., Ranganathan and Brown 2006, Peng and Gala 2014).

For each application, we measure an integration ratio defined as the share of sites (weighted by employees) that integrate a specific application (i.e., installed as part of an EMS) relative to the total number of sites that use this specific application (either as a stand-alone application or integrated into the EMS). This is akin to defining an application-specific EMS integration variable and scaling it by the general importance of the application for a given firm. Doing so allows us to focus on the specific effect of the UTSA on firms' decisions to integrate information into their EMS, relative to general, firm-specific trends in software applications.

If stronger trade secret protection incentivizes firms to generally invest in information systems, we expect the implementation of all applications to increase, regardless of whether they are stand-alone or integrated into the EMS. By contrast, if stronger trade secret protection reduces leakage costs and thus, specifically incentivizes information *integration*, we expect that the use of integrated software applications increases disproportionately to nonintegrated, stand-alone software applications. We re-estimate Equation (1) for each of our three software integration ratios (ACC, HR, and CRM). Consistent with our hypothesis and main result, $UTSA$ is positively related to the integration ratios across all three applications (Table 7). This also implies that any potential alternative mechanisms would need to specifically affect the degree of *information integration* relative to firm-specific trends in the use of software applications.

5.2.2. Financial Reporting Incentives. One specific concern might be that the UTSA impacts firms' external reporting incentives, stimulating them to increase internal information integration. This channel is, however, not

Table 7. Alternative Channels

	(1) <i>ACC-Application integration ratio</i>	(2) <i>CRM-Application integration ratio</i>	(3) <i>HR-Application integration ratio</i>
UTSA and investments in information systems			
<i>UTSA</i>	0.034*** (3.05)	0.055** (2.24)	0.055*** (2.68)
Time-varying firm and state controls	Yes	Yes	Yes
Observations	13,450	10,223	11,058
Firm FE	Yes	Yes	Yes
Industry × year FE	Yes	Yes	Yes
Within R^2	0.004	0.006	0.009
Adjusted R^2	0.676	0.759	0.719

Notes. This table reports estimates of the effect of *UTSA* on firms’ investments in other information systems. In columns (1)–(3), the dependent variables *ACC-Application integration ratio*, *CRM-Application integration ratio*, and *HR-Application integration ratios* measure the percentages of accounting, CRM, and HR applications, respectively, that are integrated (i.e., part of an EMS) relative to the overall prevalence of the respective application (applications part of an EMS and stand-alone applications). The main independent variable is *UTSA*, an indicator for whether a firm’s headquarters state has enacted the *UTSA* by year t . The same set of control variables as in Table 2 is included but not reported for conciseness. All continuous control variables, except for *NCI*, are winsorized at the 1st and 99th percentiles. We report t -statistics based on standard errors clustered at the state level in parentheses. FE, fixed effect.

Two-tailed significance levels at the 5% level; *two-tailed significance levels at the 1% level.

unequivocal but rather, multifaceted. On the one hand, stronger trade secret protection might reduce firms’ external reporting incentives as the incremental proprietary costs of public disclosure are higher (Glaeser 2018, Li et al. 2018). On the other hand, whereas stronger trade secret protection incentivizes firms to be more publicly opaque with respect to their trade secrets, it could stimulate their external reporting of nonproprietary information that is unrelated to their trade secrets. This follows as firms might aim to counteract the increase in information asymmetry because of their trade secret-related opaqueness (Glaeser 2018).

To assess the importance of this channel, we estimate the controlled direct effect following Acharya et al. (2016). This approach provides us with an estimate of the effect of the *UTSA* on *EMS integration* net of the effect of an alternative channel. If the *UTSA* affects *EMS integration* only via external reporting incentives, the controlled direct effect will be insignificant. To proxy for the external reporting incentives, we use the richness of firms’ external information environment measured by analysts following Fiechter et al. (2022).

We find that the controlled direct effect of the *UTSA* on *EMS integration* is significant and almost identical in size to our baseline effect (untabulated). This effect reflects the relation between *UTSA* and *EMS integration* after purging the estimate off the effects of the analyst following on *EMS integration*. Whereas this cannot rule out that the *UTSA* also operates via channels other than leakage costs, it does not support the notion that external reporting incentives are an important mechanism through which the *UTSA* affects firms’ internal information integration.

5.2.3. Financial Constraints. Second, trade secret protection laws may increase (incumbent) firms’ profitability, alleviating financial constraints and enabling larger investments, including investments in EMSs. We note that our tests on application integration ratios (Table 7) in the previous section already mitigate this concern; if the *UTSA* generally enables firms to spend more resources on their IT, we expect a similar increase in integrated and stand-alone applications. Yet, we find that firms specifically increase the share of applications that are integrated in an EMS.

To further assess the importance of this channel, we also estimate the controlled direct effect following Acharya et al. (2016). Following Klasa et al. (2018), we measure financial constraints by return on assets. Again, we find that the controlled direct effect of the *UTSA* on *EMS integration* is significant and very similar in size to our baseline effect (untabulated). This effect reflects the relation between *UTSA* and *EMS integration* after purging the estimate of the effects return on assets on *EMS integration*. Together, our tests do not support the notion that financial constraints are an important mechanism via which the *UTSA* affects firms’ internal information integration.

6. Conclusion

Using proprietary data, we introduce a novel proxy for firms’ internal information integration: the share of sites weighted by the number of employees that are integrated into a firm’s EMS (*EMS integration*). We exploit the staggered enactment of the *UTSA* to capture shifts in the marginal costs of internal information integration. Our findings suggest that *EMS integration* increases as the expected costs of information leakage

decrease because of stronger trade secret protection under the UTSA. We further provide evidence that the UTSA-induced increase in internal information integration translates into improvements in firms' internal information quality and decision-making quality. Our findings contribute to a more comprehensive understanding of firms' internal information environment. Specifically, we shed light on how an important strategic cost factor matters for a specific input to firms' internal information environment.

There are some potential limitations relating to our data and sample period. First, the technologies used during our sample period differ from the technologies available today (e.g., machine learning, big data, cloud computing), potentially reducing the generalizability of our findings. At the same time, our sample period covers a time when EMS adoption was growing rapidly but concerns for data leakages were similarly high (Peng and Gala 2014), providing a suitable setting to examine the cost-benefit trade-off motivating our study. Second, our data do not allow us to examine potentially different levels of quality of EMS and their actual usage by managers. Nonetheless, our path analyses indicate that, on average, internal information integration via EMS improves internal information and decision-making quality. Lastly, firms might try to protect trade secrets via other mechanisms (such

as organizational design). It would be interesting for future research to investigate how these choices interact with the integration of internal information.

Acknowledgments

The authors thank Ranjani Krishnan (department editor), an anonymous associate editor, two anonymous referees, Martin Artz (discussant), Oliver Binz, Ferdinand Elfers, Guido Friebe, John Gallemore, Pujawati Gondowijoyo, Wayne Guay, Mariya Ivanova, Eva Labro, Martin Nienhaus, Caspar David Peter, Christian Peters, Hari Ramasubramanian, Jan Riepe, Anna Rohlfing-Bastian, Harm H. Schütt, Thorsten Sellhorn, Christoph Sextroh, Laurence van Lent, and Antonio Vazquez as well as workshop participants at the Frankfurt School of Finance and Management, INSEAD, LawFin Goethe University, Rice University, Singapore Management University, Stockholm School of Economics, Tilburg University, the University of Bern, Universidad Carlos III de Madrid, the 11th European Institute for Advanced Studies in Management Conference on Performance Measurement and Management Control, the European Accounting Association's 44th Annual Congress, the 2022 Annual Conference for Management Accounting Research, the 2022 Jubiläumstagung des Verbands der Hochschullehrer für Betriebswirtschaft, the 2024 Hawai'i Accounting Research Conference, and the webinar series of the European Accounting Symposium for Young Scholars for helpful comments and suggestions. Erika Bogner, Maja Knab, Inga Meringdal, Lazaros Papadopoulos, and Wentao Ren provided excellent research assistance.

Appendix. Variable Definitions

Name	Description
Dependent variables	
<i>EMS integration</i>	The share of sites, weighted by employees, integrated into a firms' enterprise management system (CI database)
<i>ACC-Application integration ratio</i>	The number of sites, weighted by employees, that integrated an accounting application into an EMS relative to the total number of sites, weighted by employees, that have installed any accounting application (CI database)
<i>CRM-Application integration ratio</i>	The number of sites, weighted by employees, that integrated a CRM application into an EMS relative to the number of sites, weighted by employees, that have installed any CRM application (CI database)
<i>HR-Application integration ratio</i>	The number of sites, weighted by employees, that integrated an HR application into an EMS relative to the number of sites, weighted by employees, that have installed any HR application (CI database)
Independent variable	
<i>UTSA</i>	Indicator variable that equals one if the firm is headquartered in a state that enacted the UTSA and zero otherwise
$UTSA^{-4}$	Indicator variable that equals one if the firm is headquartered in a state that will enact the UTSA in 4 years and zero otherwise
$UTSA^{-3}$	Indicator variable that equals one if the firm is headquartered in a state that will enact the UTSA in 3 years and zero otherwise
$UTSA^{-2}$	Indicator variable that equals one if the firm is headquartered in a state that will enact the UTSA in 2 years and zero otherwise
$UTSA^0$	Indicator variable that equals one if the firm is headquartered in a state that enacted the UTSA in the current year and zero otherwise
$UTSA^{+1}$	Indicator variable that equals one if the firm is headquartered in a state that enacted the UTSA 1 year ago and zero otherwise
$UTSA^{+2}$	Indicator variable that equals one if the firm is headquartered in a state that enacted the UTSA 2 years ago and zero otherwise
$UTSA^{+3}$	Indicator variable that equals one if the firm is headquartered in a state that enacted the UTSA 3 years ago and zero otherwise

Appendix. (Continued)

Name	Description
<i>UTSA</i> ⁺⁴⁺	Indicator variable that equals one if the firm is headquartered in a state that enacted the UTSA 4 or more years ago and zero otherwise
Cross-sectional variables	
<i>Competition</i>	Indicator variable that equals one if competition is higher than the median, where competition is defined as the number of occurrences of competition-related words per 1,000 total words in 10-K based on Li et al. (2013)
<i>TPP</i>	Indicator variable that equals one if technological peer pressure is higher than the median and zero otherwise, where technological peer pressure is defined based on Cao et al. (2018)
<i>High-tech</i>	Indicator variable that equals one if a firm belongs to a high-tech industry according to Francis and Schipper (1999) and zero otherwise (Compustat)
<i>Vertical integration</i>	Indicator variable that equals one if vertical integration is higher than the median and zero otherwise, where vertical integration measures the degree to which a firm offers products and services that are vertically related based on Frésard et al. (2020)
Control variables	
$\log(1 + \text{Sales})$	Natural logarithm of one plus sales (sale; Compustat)
$\log(1 + \#\text{Employees})$	Natural logarithm of one plus the number of employees (emp; Compustat)
$\log(1 + \#\text{Sites})$	Natural logarithm of one plus the number of sites (CI database)
<i>PCs/employee</i>	Number of PCs scaled by number of employees (CI database)
<i>Uncertainty</i>	Environmental uncertainty calculated as the standard deviation of firm <i>i</i> 's net income (ib) for the previous 5 years, scaled by sales (sale; Compustat)
<i>Unrelated diversification</i>	Unrelated firm diversification measure of the extent to which a firm operates across 4-digit Standard Industry Classification codes (Dewan et al. 1998) (CI database)
<i>Profit</i>	Operating income (oibdp) scaled by sales (sale; Compustat)
<i>Debt ratio</i>	Long-term debt (dllt) divided by total assets (at; Compustat)
<i>Avg. sales growth</i>	Growth in sales (sale) averaged over the current and previous year (Compustat)
<i>Big 4</i>	Indicator variable that equals one if the company is audited by a Big 4 audit firm (au) and zero otherwise (Compustat)
<i>Data Breach</i>	Indicator variable that equals one if the firm is headquartered in a state that has enacted the Data Breach Notification Law and zero otherwise
<i>IDD</i>	Indicator variable that equals one if the firm is headquartered in a state that adopted the IDD and zero otherwise
<i>NCI</i>	Noncompetition enforcement index for the state in which the firm is headquartered based on Garmaise (2011)
<i>GDP growth</i>	GDP growth of the state in which the firm is headquartered (U.S. Bureau of Labor Statistic)
<i>Unemployment rate</i>	The unemployment rate of the state in which the firm is headquartered (U.S. Bureau of Labor Statistic)
<i>IT employees</i>	The percentage of information technology employees (North American Industry Classification System 5415) of the state in which the firm is headquartered (U.S. Bureau of Labor Statistic)
IIQ variables	
<i>Collaboration</i>	Share of words in the firm's 10-K filing that relate to collaboration, following Chen et al. (2022) multiplied by 100
<i>EA speed</i>	Number of days between the end of the fiscal year and the firm's earnings announcement divided by 365 and multiplied by (−1; IBES)
<i>Accrual quality</i>	The absolute value of abnormal accruals calculated using the Dechow and Dichev (2002) model and multiplied by (−1; Compustat)
<i>No error restatement</i>	Absence of unintentional restatements because of errors, following Hennes et al. (2008) and Gallemore and Labro (2015) (Audit Analytics)
DMQ variables	
<i>Capex efficiency</i>	Indicator variable that equals one if the firms' capex efficiency is above the median, calculated following Goodman et al. (2014) (Compustat)
<i>Labor efficiency</i>	Indicator variable that equals one if the firms' labor efficiency is above the median, calculated following Jung et al. (2014) (Compustat)
<i>TFP</i>	Total factor productivity, following Imrohoroglu and Tuzel (2014)

Endnotes

¹ Publicly traded U.S. firms own approximately \$5 trillion in trade secrets (U.S. Chamber of Commerce 2016), and trade secret theft is estimated to range from 1% to 3% of GDP, amounting to \$180–\$540 billion per year in the United States (The Commission on the Theft of American Intellectual Property 2017).

² A site is a distinct business location within a firm, similar to a facility or business unit.

³ In line with this, prior research finds that employees are the most frequently alleged misappropriators of trade secrets in court trials (Almeling et al. 2010).

⁴ Prior literature finds that EMSs enhance risk management (Tian and Xu 2015), increase the relevance of internal signals for investment decisions (Heitzman and Huang 2019), foster innovation (Forman and Van Zeebroeck 2012, Kleis et al. 2012, Huang et al. 2020), improve firms' external reporting (Brazel and Dang 2008, Morris 2011, Dorantes et al. 2013, Pincus et al. 2017), and can ultimately improve firm

performance (Anderson et al. 2006, Ranganathan and Brown 2006, Hendricks et al. 2007, Setia et al. 2011, Li and Sandino 2018, Nagle 2019).

⁵ In 2018, the National Cybersecurity and Communications Integration Center of Homeland Security issued an alert on cyberattacks because “firms store highly sensitive data—including financial results, manufacturing secrets, and credit card numbers—in ... enterprise resource planning (ERP) software” (Auchard 2018). In line with this, (former) employees have been accused of downloading confidential information from firm’s information systems in UTSA litigation cases. For instance, in *Engility Corp. v. Daniels* (No. 16-cv-2473-WJM-MEH, 2016 WL 7034976, at *6, *8; D. Colo. Dec. 2, 2016), former employee Daniels was accused of stealing information on product pricing, profitability information, customer contacts, product designs, etc. with a flash drive. In *Xantrex Tech. Inc. v. Advance Energy Indus., Inc.* (2008 WL 2185882, at *18 (D. Colo. 2008)), former employee Thompson was accused of accessing highly confidential and trade secret competitive files and either saving them to an external source or emailing them to his and his wife’s personal email accounts.

⁶ We take information about the location of a firm’s headquarters state from the Computer Intelligence Database (CI database), which records the headquarters state in yearly (historical) snapshots.

⁷ For example, in *U.S. Gypsum Co. v. LaFarge N. Am., Inc.* (508 F. Supp. 2d 601 N.D. Ill. 2007), none of the defendants have ever lived in Illinois nor been employed in a position where their place of employment was Illinois. However, all are former employees of U.S. Gypsum, headquartered in Illinois. Citing Seventh Circuit precedent, the court found personal jurisdiction because the defendants were aware that U.S. Gypsum was an Illinois-based company when they allegedly misappropriated its trade secrets.

⁸ If not all sites of a firm were interviewed in a given year, we follow Nagle (2019) and assign the average values of the interviewed sites to the noninterviewed sites to construct our firm-year-level variables. Ninety percent of the sites in our sample are interviewed every year.

⁹ The IDD reduces employee mobility by stating that a former employee can be prevented from working for a competitor (e.g., Klasa et al. 2018, Li et al. 2018). Notably, decreased employee mobility might affect firms’ decisions to integrate internal information for reasons other than leakage costs, such as increased employee tenure (Labro and Omartian 2021). Thus, we include IDD adoption as a control variable in our empirical tests. Our inferences remain the same if we also consider IDD rejections.

¹⁰ We address concerns relating to confounding events and a small number of treatment states in Section 5.1.

¹¹ The correlation between these two measures is high ($\rho = 0.90$). In Section 5.2, we show that our findings are robust to alternative definitions of our dependent variable, including assigning equal weights to each site and accounting for site-level characteristics.

¹² We further address time-varying treatment effects in the presence of treatment heterogeneity in DID estimation using the “stacked” regression following Cengiz et al. (2019). Untabulated results suggest that the aforementioned bias is not a concern in our setting.

¹³ Our inferences remain unchanged when we use less granular fixed effects (i.e., state and industry-year fixed effects; untabulated), reducing the likelihood of a spurious correlation (Jennings et al. 2023).

¹⁴ Similarly, our results are unchanged when we constrain our sample to a balanced panel of firms for which we have data in all years ($N = 8,690$), reducing the concern that our results could be influenced by structural changes in the data collection process.

¹⁵ Here, $22.96\% = 0.045/0.2003$, where 0.2003 is the within-fixed effects standard deviation of *EMS integration* (deHaan 2021).

¹⁶ Corroborating the notion that our cross-sectional proxies adequately capture firms’ marginal costs/benefits of *EMS integration*, we

find (untabulated) that firms with higher external competition, firms with higher technological peer pressure, and firms belonging to the high-tech industry exhibit, on average, lower levels of *EMS integration*. Conversely, firms with high vertical integration exhibit, on average, lower levels of *EMS integration*.

¹⁷ The only exception is the analysis of accrual quality. Following Dechow and Dichev (2002), we measure accrual quality as the within-firm standard deviation of abnormal accruals over multiple years. Thus, this measure exhibits little variation within firm over time, rendering firm fixed effects infeasible. Instead, we use industry $\times$ year and state fixed effects.

¹⁸ Similarly, our results are unaffected if we include separate indicators per treated state in one regression. To address the concern of few treatments in our sample period (Conley and Taber 2011), we also implement the wild bootstrap randomization inference test (MacKinnon and Webb 2019), which accounts for few treatments (see Conley and Taber 2011) and few control groups. Our inferences remain unchanged, with all p -values for coefficients remaining significant at the 10% level or better (untabulated).

¹⁹ As this approach leads to fewer clusters, we calculate confidence intervals using the wild-bootstrap method (Cameron et al. 2008).

²⁰ Bloom et al. (2014) document that EMS implementation is not associated with greater centralization or monitoring but rather, that it improves local managers’ information access.

²¹ We do not have sufficient observations to run these tests with the supply chain-management application.

References

- Abernethy M, Bouwens J (2005) Determinants of accounting innovation implementation. *Abacus* 41(3):217–240.
- Aboody D, Lev B (2000) Information asymmetry, R&D, and insider gains. *J. Finance* 55(6):2747–2766.
- Acharya A, Blackwell M, Sen M (2016) Explaining causal findings without bias: Detecting and assessing direct effects. *Amer. Political Sci. Rev.* 110(3):512–529.
- Ahmad A, Bosua R, Scheepers R (2014) Protecting organizational competitive advantage: A knowledge leakage perspective. *Comput. Security* 42(May):27–39.
- Almeling DS, Snyder DW, Sapoznikow M, McColluman WE, Weader J (2010) A statistical analysis of trade secret litigation in state courts. *Gonzaga Law Rev.* 46(1):57–101.
- Amiraslani H, Deller C, Ittner C, Keusch T (2024) The importance of directors’ information access: Evidence from board risk reporting. Working paper, INSEAD, Fontainebleau, France.
- Anderson MC, Banker RD, Ravindran S (2006) Value implications of investments in information technology. *Management Sci.* 52(9):1359–1376.
- Anderson SW, Christ MH, Dekker HC, Sedatole KL (2014) The use of management controls to mitigate risk in strategic alliances: Field and survey evidence. *J. Management Accounting Res.* 26(1):1–32.
- Arundel A (2001) The relative effectiveness of patents and secrecy for appropriation. *Res. Policy* 30(4):611–624.
- Asker J, Ljungqvist A (2010) Competition and the structure of vertical relationships in capital markets. *J. Political Econom.* 118(3):599–647.
- Auchard E (2018) Study warns of cyber hacking risk to firms using SAP, Oracle management software. *Insurance J.* (July 25), <https://www.insurancejournal.com/news/national/2018/07/25/495977.htm>.
- Baldwin CY, Henkel J (2015) Modularity and intellectual property protection. *Strategic Management J.* 36(11):1637–1655.
- Barrios JM (2022) Occupational licensing and accountant quality: Evidence from the 150-hour rule. *J. Accounting Res.* 60(1):3–43.
- Beaudry P, Doms M, Lewis E (2010) Should the personal computer be considered a technological revolution? Evidence from U.S. metropolitan areas. *J. Political Econom.* 118(5):988–1036.

- Bens DA, Berger PG, Monahan SJ (2011) Discretionary disclosure in financial reporting: An examination comparing internal firm data to externally reported segment data. *Accounting Rev.* 86(2):417–449.
- Berger PG, Hann RN (2007) Segment profitability and the proprietary and agency costs of disclosure. *Accounting Rev.* 82(4):869–906.
- Berman DL, Otterberg AA, Maleson JA, Salander AM, Hersch AJ (2017) Understanding and litigating trade secrets. Accessed February 22, 2023, https://www.jenner.com/a/web/ejymw5kQ4aThswmb7tntdh/4HRMZQ/Trade_Secrets_Outline_0617_PRINT.PDF?1497540235.
- Bills KL, Cobabe M, Pittman J, Stein SE (2020) To share or not to share: The importance of peer firm similarity to auditor choice. *Accounting Organ. Soc.* 83(May):101115.
- Bloom N, Garicano L, Sadun R, Van Reenen J (2014) The distinct effects of information technology and communication technology on firm organization. *Management Sci.* 60(12):2859–2885.
- Bouwens J, Abernethy MA (2000) The consequences of customization on management accounting system design. *Accounting Organ. Soc.* 25(3):221–241.
- Brazel JF, Dang L (2008) The effect of ERP system implementations on the management of earnings and earnings release dates. *J. Inform. Systems* 22(2):1–21.
- Busenbark JR, Yoon H, Gamache DL, Withers MC (2022) Omitted variable bias: Examining management research with the impact threshold of a confounding variable (ITCV). *J. Management* 48(1):17–48.
- Cabazon F, Hoberg G (2022) Leaky director networks and innovation herding. Working paper, Virginia Tech Pamplin College of Business, Blacksburg, VA.
- Cameron A, Gelbach J, Miller D (2008) Bootstrap-based improvements for inference with clustered errors. *Rev. Econom. Statist.* 90(3):414–427.
- Cao SS, Ma G, Tucker JW, Wan C (2018) Technological peer pressure and product disclosure. *Accounting Rev.* 93(6):95–126.
- Cassar G, Gibson B (2008) Budgets, internal reports, and manager forecast accuracy. *Contemporary Accounting Res.* 25(3):707–738.
- Castellaneta F, Conti R, Kacperczyk A (2017) Money secrets: How does trade secret legal protection affect firm market value? Evidence from the uniform trade secret act. *Strategic Management J.* 38(4):834–853.
- Cengiz D, Dube A, Lindner A, Zipperer B (2019) the effect of minimum wages on low-wage jobs. *Quart. J. Econom.* 134(3):1405–1454.
- Chen X, Hilary G, Tian X (2019) Data breach disclosure and insider trading. Working paper, McDonough School of Business, Georgetown University, Washington, DC.
- Chen HM, Francis BB, Hasan T, Wu Q (2022) Does Corporate culture impact audit pricing? Evidence from textual analysis. *J. Bus. Finance Accounting* 49(5–6):778–806.
- Chen C, Martin X, Roychowdhury S, Wang X, Billett MT (2018) Clarity begins at home: Internal information asymmetry and external communication quality. *Accounting Rev.* 93(1):71–101.
- Cheng Z, Rai A, Tian F, Xu SX (2021) Social learning in information technology investment: The role of board interlocks. *Management Sci.* 67(1):547–576.
- Cheng L, Li Y, Li W, Holm E, Zhai Q (2013) Understanding the violation of IS security policy in organizations: An integrated model based on social control and deterrence theory. *Comput. Security* 39(November):447–459.
- Chenhall RH, Morris D (1986) The impact of structure, environment, and interdependence on the perceived usefulness of management accounting systems. *Accounting Rev.* 61(1):16–35.
- Christ MH, Nicolaou AI (2016) Integrated information systems, alliance formation, and the risk of information exchange between partners. *J. Management Accounting Res.* 28(3):1–18.
- Clemons EK, Hitt LM (2004) Poaching and the misappropriation of information: Transaction risks of information exchange. *J. Management Inform. Systems* 21(2):87–107.
- Cohen W, Nelson R, Walsh J (2000) Protecting their intellectual assets: Appropriability conditions and why U.S. manufacturing firms patent (or not). NBER working paper, National Bureau of Economic Research, Cambridge, MA.
- Conley T, Taber CR (2011) Inference with “difference in differences” with a small number of policy changes. *Rev. Econom. Statist.* 93(1):113–125.
- Contigiani A, Hsu DH, Barankay I (2018) Trade secrets and innovation: Evidence from the “inevitable disclosure” doctrine. *Strategic Management J.* 39(11):2921–2942.
- Davila A, Foster G (2005) Management accounting systems adoption decisions: Evidence and performance implications from early-stage startup companies. *Accounting Rev.* 80(4):1039–1068.
- Dechow PM, Dichev ID (2002) The quality of accruals and earnings: The role of accrual estimation errors. *Accounting Rev.* 77(1):35–59.
- deHaan E (2021) Using and interpreting fixed effects models. Working paper, Stanford Graduate School of Business, Stanford, CA.
- Dewan S, Michael SC, Min C-k (1998) Firm characteristics and investments in information technology: Scale and scope effects. *Inform. Systems Res.* 9(3):219–232.
- Dorantes C-A, Li C, Peters GF, Richardson VJ (2013) The effect of enterprise systems implementation on the firm information environment. *Contemporary Accounting Res.* 30(4):1427–1461.
- Drake MS, Hess RV, Wilde JH, Williams BM (2022) The relevance of non-income tax relief. *Contemporary Accounting Res.* 39(3):1797–1833.
- Eisenhardt KM, Martin JA (2000) Dynamic capabilities: What are they? *Strategic Management J.* 21(10/11):1105–1121.
- Elliott K, Pataconi A, Swierzbinski J, Williams J (2019) Knowledge protection in firms: A conceptual framework and evidence from HP Labs. *Eur. Management Rev.* 16(1):179–193.
- Fiechter P, Hitz JM, Lehmann N (2022) Real effects of a widespread CSR reporting mandate: Evidence from the European Union’s CSR Directive. *J. Accounting Res.* 60(4):1499–1549.
- Forman C (2005) The corporate digital divide: Determinants of internet adoption. *Management Sci.* 51(4):641–654.
- Forman C, Van Zeebroeck N (2012) From wires to partners: How the internet has fostered R&D collaborations within firms. *Management Sci.* 58(8):1549–1568.
- Francis J, Schipper K (1999) Have financial statements lost their relevance? *J. Accounting Res.* 37(2):319–352.
- Frank KA (2000) Impact of a confounding variable on a regression coefficient. *Sociol. Methods Res.* 29(2):147–194.
- Franzoni LA, Kaushik A (2016) The optimal scope of trade secrets law. *Internat. Rev. Law Econom.* 45(C):45–53.
- Frésard L, Hoberg G, Phillips GM (2020) Innovation activities and integration through vertical acquisitions. *Rev. Financial Stud.* 33(7):2937–2976.
- Gallemore J, Labro E (2015) The importance of the internal information environment for tax avoidance. *J. Accounting Econom.* 60(1):149–167.
- Garmaise MJ (2011) Ties that truly bind: Noncompetition agreements, executive compensation, and firm investment. *J. Law Econom. Organ.* 27(2):376–425.
- Glaeser S (2018) The effects of proprietary information on corporate disclosure and transparency: Evidence from trade secrets. *J. Accounting Econom.* 66(1):163–193.
- Glaeser CK, Glaeser S, Labro E (2023) Proximity and the management of innovation. *Management Sci.* 69(5):3080–3099.
- Godigbe BG, Jennings J, Seo H, Tanlu L (2022) The effect of geographic diversity on managerial earnings forecasts. *Eur. Accounting Rev.*, ePub ahead of print November 4, <https://doi.org/10.1080/09638180.2022.2139738>.
- Goodman TH, Neamtui M, Shroff N, White HD (2014) Management forecast quality and capital investment decisions. *Accounting Rev.* 89(1):331–365.

- Gormley TA, Matsa DA (2014) Common errors: How to (and not to) control for unobserved heterogeneity. *Rev. Financial Stud.* 27(2): 617–661.
- Granlund M, Malmi T (2002) Moderate impact of ERPS on management accounting: A lag or permanent outcome? *Management Accounting Res.* 13(3):299–321.
- Ham JC, Ueda K (2021) The employment impact of the provision of public health insurance: A further examination of the effect of the 2005 TennCare contraction. *J. Labor Econom.* 39(S1):S199–S238.
- Hannah D (2005) Should I keep a secret? The effects of trade secret protection procedures on employees' obligations to protect trade secrets. *Organ. Sci.* 16(1):71–84.
- Heitzman S, Huang M (2019) Internal information quality and the sensitivity of investment to market prices and accounting profits. *Contemporary Accounting Res.* 36(3):1699–1723.
- Hellmann T, Perotti E (2011) The circulation of ideas in firms and markets. *Management Sci.* 57(10):1813–1826.
- Hendricks KB, Singhal VR, Stratman JK (2007) The impact of enterprise systems on corporate performance: A study of ERP, SCM, and CRM system implementations. *J. Oper. Management* 25(1):65–82.
- Hennes KM, Leone AJ, Miller BP (2008) The importance of distinguishing errors from irregularities in restatement research: The case of restatements and CEO/CFO. *Accounting Rev.* 83(6):1487–1519.
- Huang K, Lao B, McPhee G (2020) Internal information quality and patent-related innovation. *J. Bus. Finance Accounting* 47(3–4):489–518.
- Huth CL, Chadwick DW, Claycomb WR, You I (2013) Guest editorial: A brief overview of data leakage and insider threats. *Inform. Systems Frontiers* 15(1):1–4.
- Imrohorglu A, Tuzel S (2014) Firm-level productivity, risk, and return. *Management Sci.* 60(8):2073–2090.
- InfoWatch (2006) Internal IT threats in Europe 2006. Accessed February 22, 2023, <https://securelist.com/internal-it-threats-in-europe-2006/36142/>.
- InfoWatch (2014) Global Data Leakage Report 2013. Accessed February 22, 2023, <https://infowatch.com/analytics/reports/3641>.
- InfoWatch (2018) Insiders compromise three times more data in H1 2018. Accessed April 20, 2024, <https://infowatch.com/news/insiders-compromised-three-times-more-data-first-half-two-thousand-eighteen>.
- Ittner CD, Lanen WN, Larcker DF (2002) The association between activity-based costing and manufacturing performance. *J. Accounting Res.* 40(3):711–726.
- Jennings JN, Kim JM, Lee JA, Taylor D (2023) Measurement error, fixed effects, and false positives in accounting research. *Rev. Accounting Stud.*, ePub ahead of print March 14, <https://doi.org/10.1007/s11142-023-09754-z>.
- Jones H (2014) Pitfalls of choice of law clauses in employment contracts. Accessed February 22, 2023, <https://hr.blr.com/HR-news/Staffing-Training/Employment-Contracts/Pitfalls-of-choice-of-law-clauses-in-employment-co/>.
- Jung B, Lee WJ, Weber DP (2014) Financial reporting quality and labor investment efficiency. *Contemporary Accounting Res.* 31(4):1047–1076.
- Kamiya S, Kang JK, Kim J, Milidonis A, Stulz RM (2021) Risk management, firm reputation, and the impact of successful cyberattacks on target firms. *J. Financial Econom.* 139(3):719–749.
- Kim Y, Su L, Wang Z, Wu H (2021) The effect of trade secrets law on stock price synchronicity: Evidence from the inevitable disclosure doctrine. *Accounting Rev.* 96(1):325–348.
- Klasa S, Ortiz-Molina H, Serfling M, Srinivasan S (2018) Protection of trade secrets and capital structure decisions. *J. Financial Econom.* 128(2):266–286.
- Kleis L, Chwelos P, Ramirez RV, Cockburn I (2012) Information technology and intangible output: The impact of IT investment on innovation productivity. *Inform. Systems Res.* 23(1):42–59.
- Labro E, Omartian JD (2021) Managing employee retention concerns: Evidence from US Census data. Working paper, University of North Carolina at Chapel Hill, Chapel Hill, NC.
- Labro E, Stice-Lawrence L (2020) Updating accounting systems: Longitudinal evidence from the healthcare sector. *Management Sci.* 66(12):6042–6061.
- Labro E, Peter CD, Pierk J, Van Linden C (2022) In-network auditors as information intermediaries within business groups. Working paper, University of North Carolina at Chapel Hill, Chapel Hill, NC.
- Larcker DF, Rusticus TO (2010) On the use of instrumental variables in accounting research. *J. Accounting Econom.* 49(3):186–205.
- Li Y, Li Y (2020) The effect of trade secrets protection on disclosure of forward-looking financial information. *J. Bus. Finance Accounting* 47(3–4):397–437.
- Li SX, Sandino T (2018) Effects of an information sharing system on employee creativity, engagement, and performance. *J. Accounting Res.* 56(2):713–747.
- Li Y, Lin Y, Zhang L (2018) Trade secrets law and corporate disclosure: Causal evidence on the proprietary cost hypothesis. *J. Accounting Res.* 56(1):265–308.
- Li F, Lundholm R, Minnis M (2013) A measure of competition based on 10-K filings. *J. Accounting Res.* 51(2):399–436.
- Liebeskind JP (1996) Knowledge, strategy, and the theory of the firm: Knowledge, strategy, and the theory of the firm. *Strategic Management J.* 17(S2):93–107.
- Liebeskind JP (1997) Keeping organizational secrets: Protective institutional mechanisms and their costs. *Indust. Corporate Change* 6(3):623–663.
- Lyle MR (2019) Information quality, growth options, and average future stock returns. *Accounting Rev.* 94(1):271–298.
- MacKinnon JG, Webb MD (2019) Wild bootstrap randomization inference for few treated clusters. Huynh KP, Jacho-Chávez DT, Tripathi G, eds. *The Econometrics of Complex Survey Data*, vol. 39 (Emerald Publishing Limited, Leeds, UK), 61–85.
- Molok NNA, Ahmad A, Chang S (2018) A case analysis of securing organisations against information leakage through online social networking. *Internat. J. Inform. Management* 43(December): 351–356.
- Morris JJ (2011) The impact of enterprise resource planning (ERP) systems on the effectiveness of internal controls over financial reporting. *J. Inform. Systems* 25(1):129–157.
- Morris MG, Venkatesh V (2010) Job characteristics and job satisfaction: Understanding the role of enterprise resource planning system implementation. *MIS Quart.* 34(1):143–161.
- Nagle F (2019) Open source software and firm productivity. *Management Sci.* 65(3):1191–1215.
- National Science Foundation, National Center for Science and Engineering Statistics (2015) Business research and development and innovation: 2012. Working paper, National Science Foundation, Alexandria, VA.
- Peng GC, Gala CJ (2014) Cloud ERP: A new dilemma to modern organisations? *J. Comput. Inform. Systems* 54(3):22–30.
- Pierri N, Timmer Y (2020) Tech in fin before fintech: Blessing or curse for financial stability? Working paper, Center for Economic Studies and ifo Institute (CESifo), Munich, Germany.
- Pincus M, Tian F, Wellmeyer P, Xu SX (2017) Do clients' enterprise systems affect audit quality and efficiency? *Contemporary Accounting Res.* 34(4):1975–2021.
- Pinnuck M, Lillis AM (2007) Profits vs. losses: Does reporting an accounting loss act as a heuristic trigger to exercise the abandonment option and divest employees? *Accounting Rev.* 82(4): 1031–1053.
- Png I (2017a) Law and innovation: Evidence from state trade secrets laws. *Rev. Econom. Statist.* 99(1):167–179.
- Png I (2017b) Secrecy and patents: Theory and evidence from the Uniform Trade Secrets Act. *Strategy Sci.* 2(3):176–193.
- Pooley J (1997) *Trade Secrets* (Law Journal Press, New York).
- Porter JD (2016) Determining choice of law in civil litigation. Accessed February 22, 2023, <https://www.jdporterlaw.com/285-2/determining-choice-law-civil-litigation/>.

- Qiu B, Wang T (2018) Does knowledge protection benefit shareholders? Evidence from stock market reaction and firm investment in knowledge assets. *J. Financial Quant. Anal.* 53(3):1341–1370.
- Ranganathan C, Brown CV (2006) ERP investments and the market value of firms: Toward an understanding of influential ERP project variables. *Inform. Systems Res.* 17(2):145–161.
- Ribstein LE, Kobayashi BH (1996) An economic analysis of uniform state laws. *J. Legal Stud.* 25(1):131–199.
- Robertson KM, Hannah DR, Lautsch BA (2015) The secret to protecting trade secrets: How to create positive secrecy climates in organizations. *Bus. Horizons* 58(6):669–677.
- Romanosky S, Telang R, Acquisti A (2011) Do data breach disclosure laws reduce identity theft? *J. Policy Anal. Management* 30(2): 256–286.
- Ronde T (2001) Trade secrets and information sharing. *J. Econom. Management Strategy* 10(3):391–417.
- Setia P, Setia M, Krishnan R, Sambamurthy V (2011) The effects of the assimilation and use of IT applications on financial performance in healthcare organizations. *J. Assoc. Inform. Systems* 12(3):274–298.
- Shapiro RJ, Hassett KA (2005) The economic value of intellectual property. Accessed April 22, 2024, <https://www.sonecon.com/docs/studies/IntellectualPropertyReport-October2005.pdf>.
- Steinmeyer PA, Becker E, Green PC, Jackson ZC (2019) Trade secret litigation. Practical law note published by Thomson Reuters. [https://uk.practicallaw.thomsonreuters.com/5-523-8283?contextData=\(sc.Default\)&transitionType=Default&firstPage=true](https://uk.practicallaw.thomsonreuters.com/5-523-8283?contextData=(sc.Default)&transitionType=Default&firstPage=true).
- The Commission on the Theft of American Intellectual Property (2017) Update to the IP Commission Report. The theft of American intellectual property: Reassessments of the challenge and United States policy, NBER Working Paper, National Bureau of Economic Research, Cambridge, MA.
- Tian F, Xu SX (2015) How do enterprise resource planning systems affect firm risk? Post-implementation impact. *MIS Quart.* 39(1):39–60.
- U.S. Chamber of Commerce (2016) The case for enhanced protection of trade secrets in the trans-pacific partnership agreement. Working paper, U.S. Chamber of Commerce, Washington, DC.
- von Rheinbaben J, Ruckes M (2004) The number and the closeness of bank relationships. *J. Banking Finance* 28(7):1597–1615.
- Wang J, Burke K (2022) The effects of disclosure and enforcement on payday lending in Texas. *J. Financial Econom.* 145(2):489–507.
- Wong WP, Tan HC, Tan KH, Tseng M-L (2019) Human factors in information leakage: Mitigation strategies for information sharing integrity. *Indust. Management Data Systems* 119(6): 1242–1267.
- Zhao XS, Lynch JG, Chen QM (2010) Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *J. Consumer Res.* 37(2):197–206.